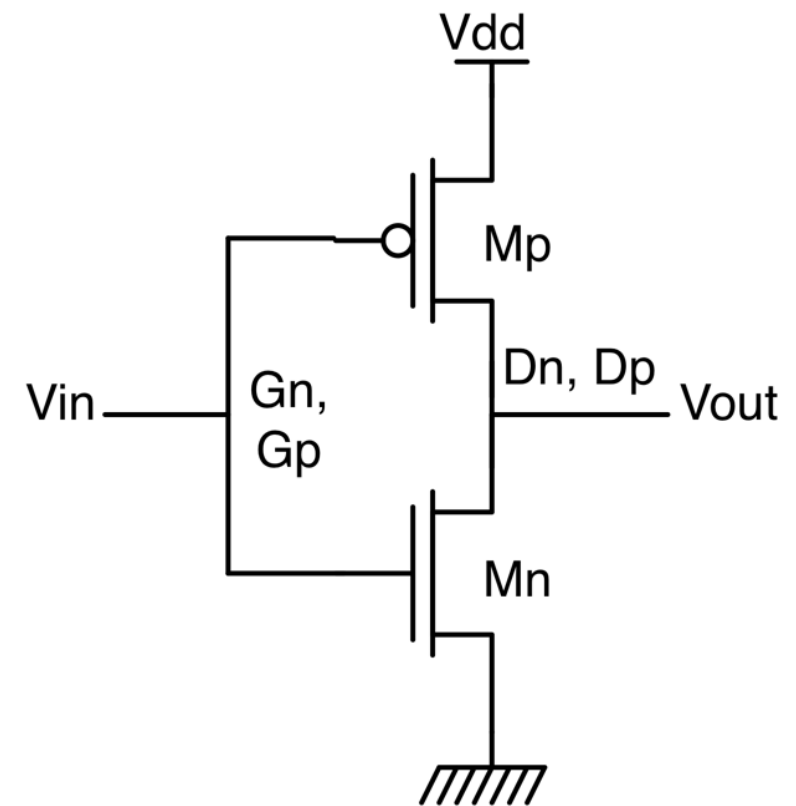


3. CMOS

STRUCTURE OF THE CMOS INVERTER

- The PMOS is a pull-up switch
- The NMOS is a pull-down switch
- The input is the gate of both
- The output is the drain of both
- Sources of both are at each of the rails



INTUITION ABOUT REGIONS OF OPERATION

- An NMOS with a gate at V_{dd} is (almost) certain to be on
- A PMOS with a gate at ground is (almost) certain to be on
- An NMOS with a drain at a very high voltage is likely to be saturated
- A PMOS with a drain at a very low voltage is likely to be saturated
- A MOSFET with a large channel field is saturated
 - A MOSFET with a large absolute value of V_{ds} is likely to be saturated

SOLVING FOR 00H

- Still can only do KCL at V_{out}
- Region of operation still makes all the difference

- $V_{in} = V_{dd}$

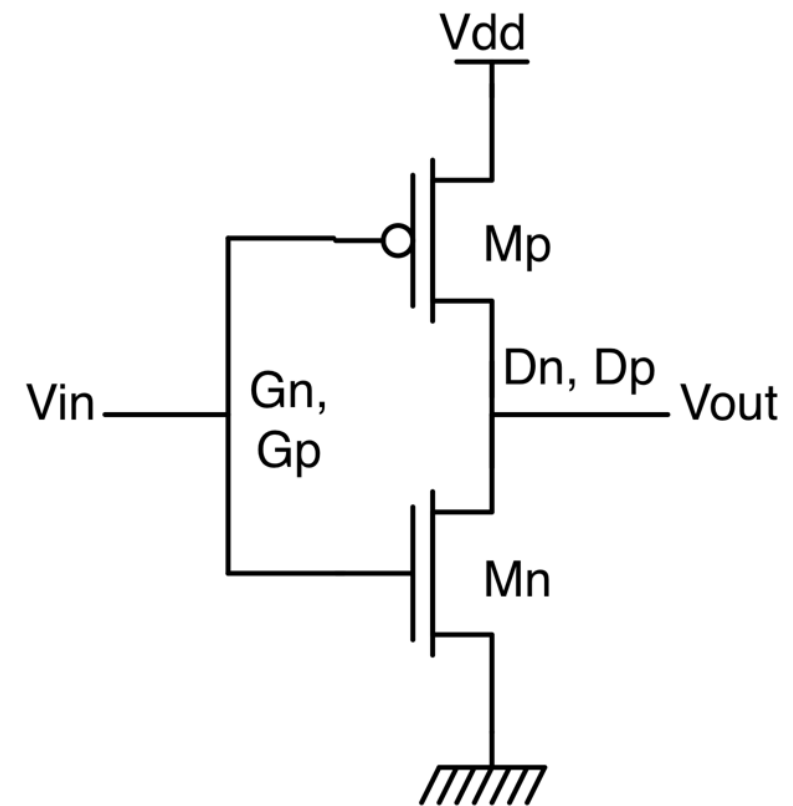
$$I_{Mn} = I_{Mp}$$

$$I_{Mn} = I_{Mp} \Big|_{cutoff} = 0$$

$$I_{Mn} \Big|_{ohmic} = 0 = K_n((V_{gs} - V_{thn})V_{ds} - 0.5V_{ds}^2)$$

$$I_{Mn} \Big|_{ohmic} = 0 = K_n((V_{DD} - V_{thn})V_{out} - 0.5V_{out}^2)$$

$$V_{out} = V_{ol} = 0$$



FINDING VOL

► $V_{in} = V_{DD}$

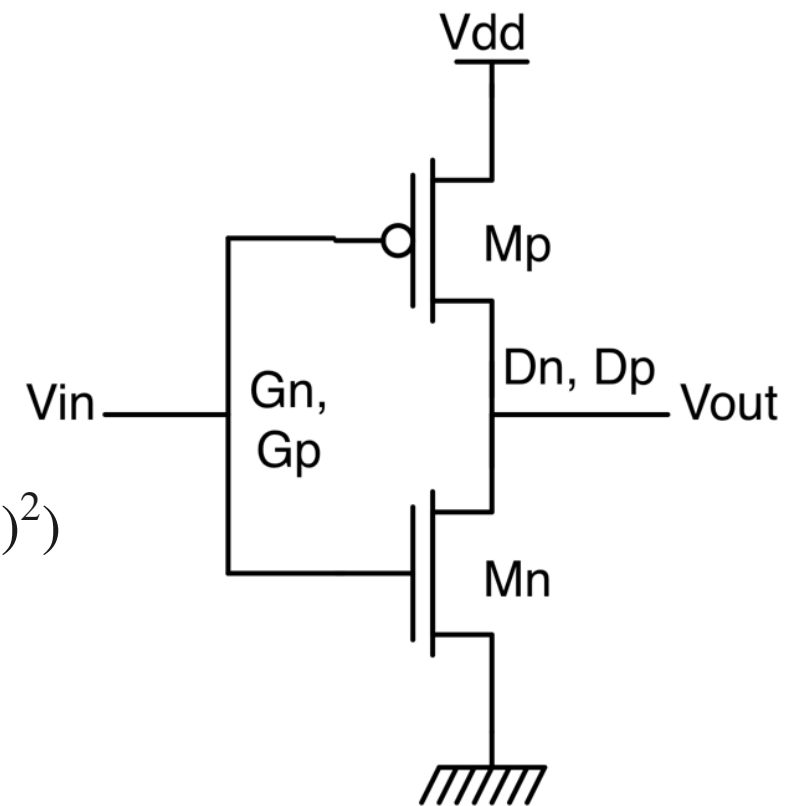
$$I_{Mp} = I_{Mn} \Big|_{cutoff} = 0$$

$$I_{Mp} \Big|_{ohmic} = 0 = K_p((V_{gs} - V_{thp})V_{ds} - 0.5V_{ds}^2)$$

$$I_{Mp} \Big|_{ohmic} = 0 = K_n((-V_{DD} - V_{thp})(V_{out} - V_{DD}) - 0.5(V_{out} - V_{DD})^2)$$

$$(V_{out} - V_{DD}) = 0$$

$$V_{out} = V_{oh} = V_{DD}$$

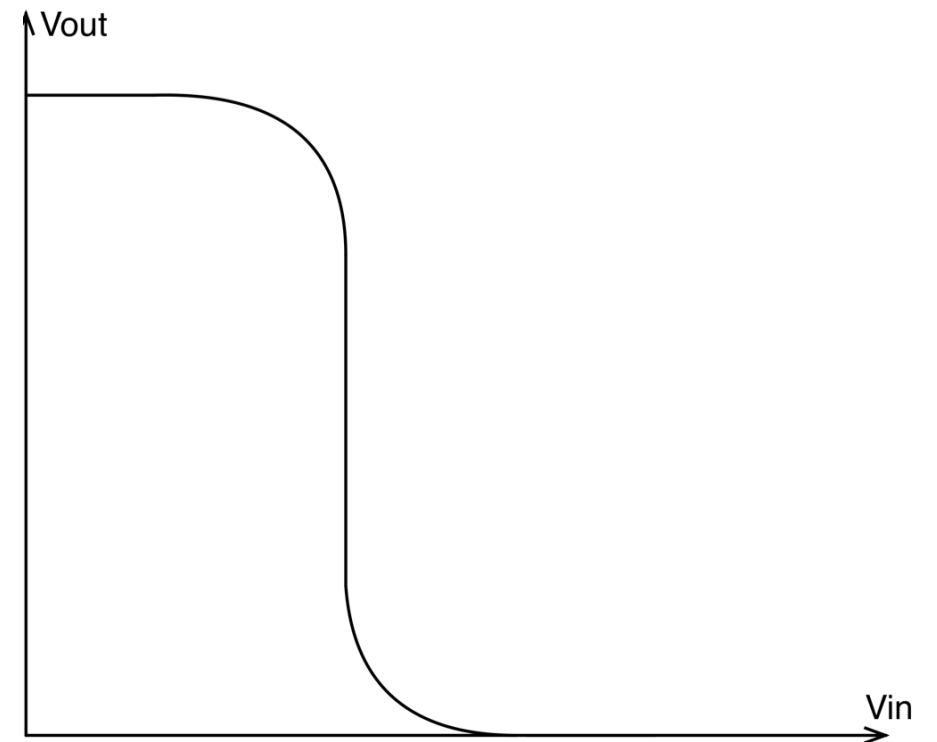


NOTES ABOUT THE CMOS INVERTER

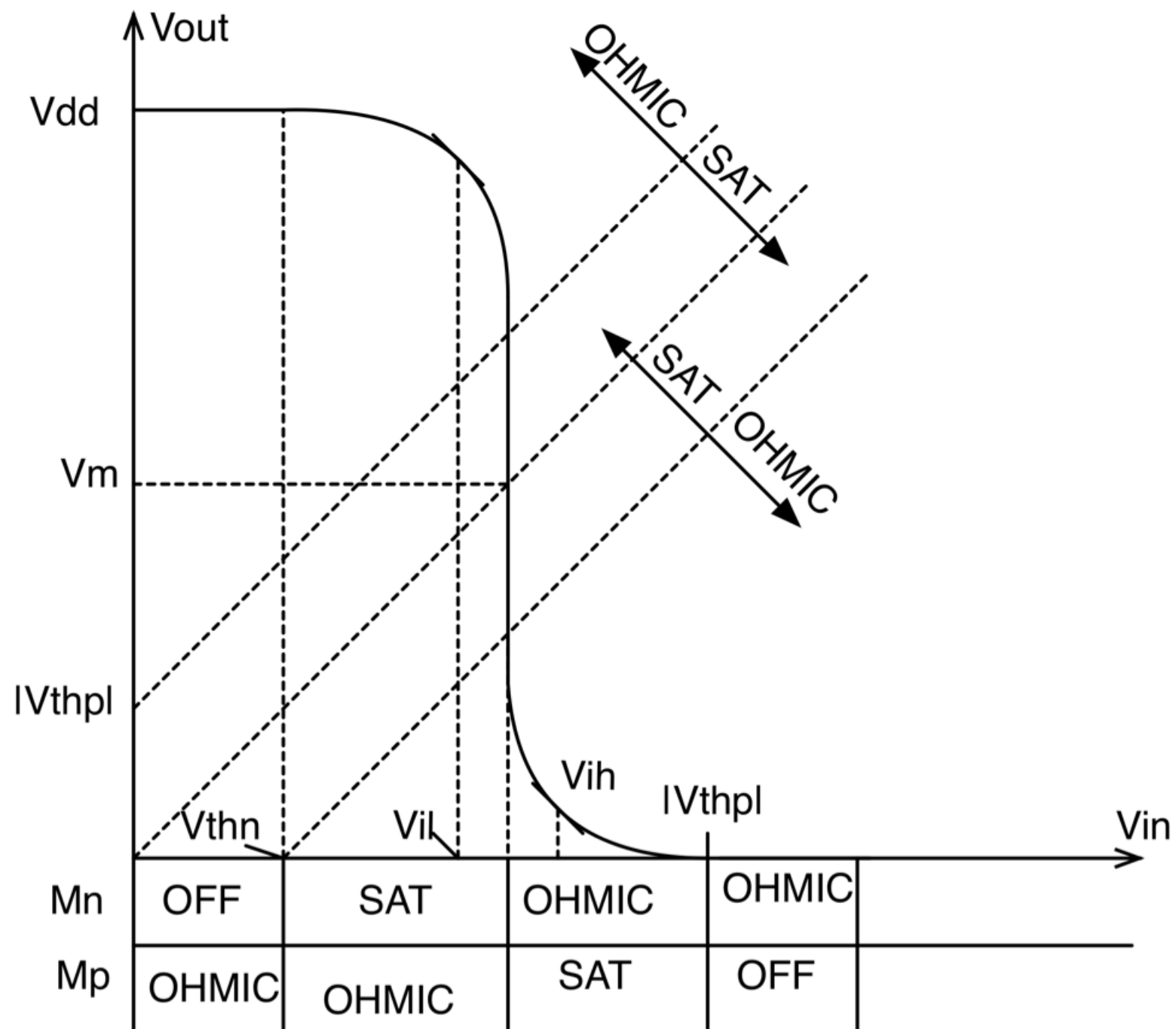
- The inverter is already starting to look pretty ideal
 - $V_{oh} = V_{dd}$ and $V_{ol} = 0V$
 - Neither outputs are a function of transistor parameters
 - For both outputs static current is null, leading to zero static power
- We only need to ensure noise margins have improved

THE VIC

- The transition region looks very sharp
- Noise margins look much improved (over ratioed logic)
- Still have to derive and prove



DERIVING THE CMOS VTC



OBTAINING VM

- The region of operation is definitely sat-sat (why)?

$$I_{Mn} \Big|_{sat} = I_{Mp} \Big|_{sat}$$

$$K_n(V_{in} - V_{thn})^2 = K_p(V_{in} - V_{DD} - V_{thp})^2$$

$$K_n(V_m - V_{thn})^2 = K_p(V_m - V_{DD} - V_{thp})^2$$

$$\sqrt{K_n/K_p}(V_m - V_{thn}) = -(V_m - V_{DD} - V_{thp})$$

$$V_m = \frac{V_{DD} + V_{thn}\sqrt{K_n/K_p} + V_{thp}}{1 + \sqrt{K_n/K_p}}$$

FOR SYMMETRIC NOISE

► For symmetric noise we need $V_m = V_{DD}/2$

$$V_m = \frac{V_{DD} + V_{thn}\sqrt{K_n/K_p} + V_{thp}}{1 + \sqrt{K_n/K_p}}$$

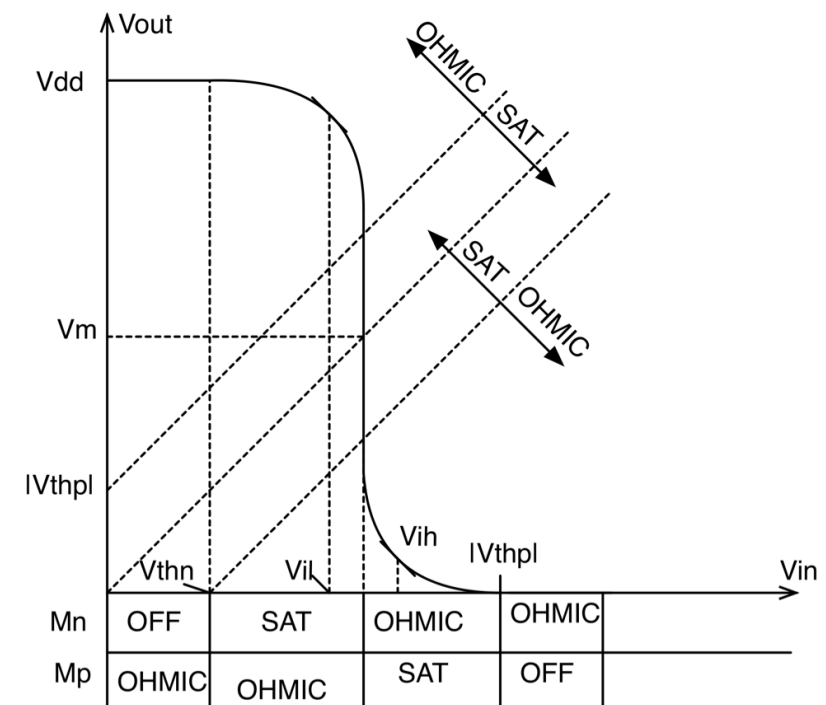
$$K_n/K_p = 1 = \frac{\mu_n(W/L)_n}{\mu_p(W/L)_p}$$

$$\frac{(W/L)_n}{(W/L)_p} = \frac{\mu_p}{\mu_n} = \frac{1}{2}$$

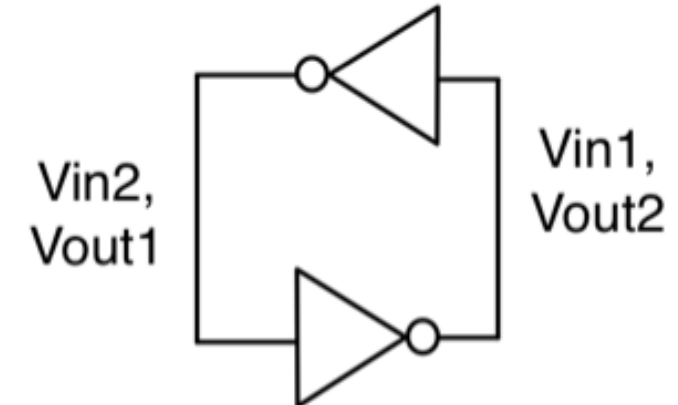
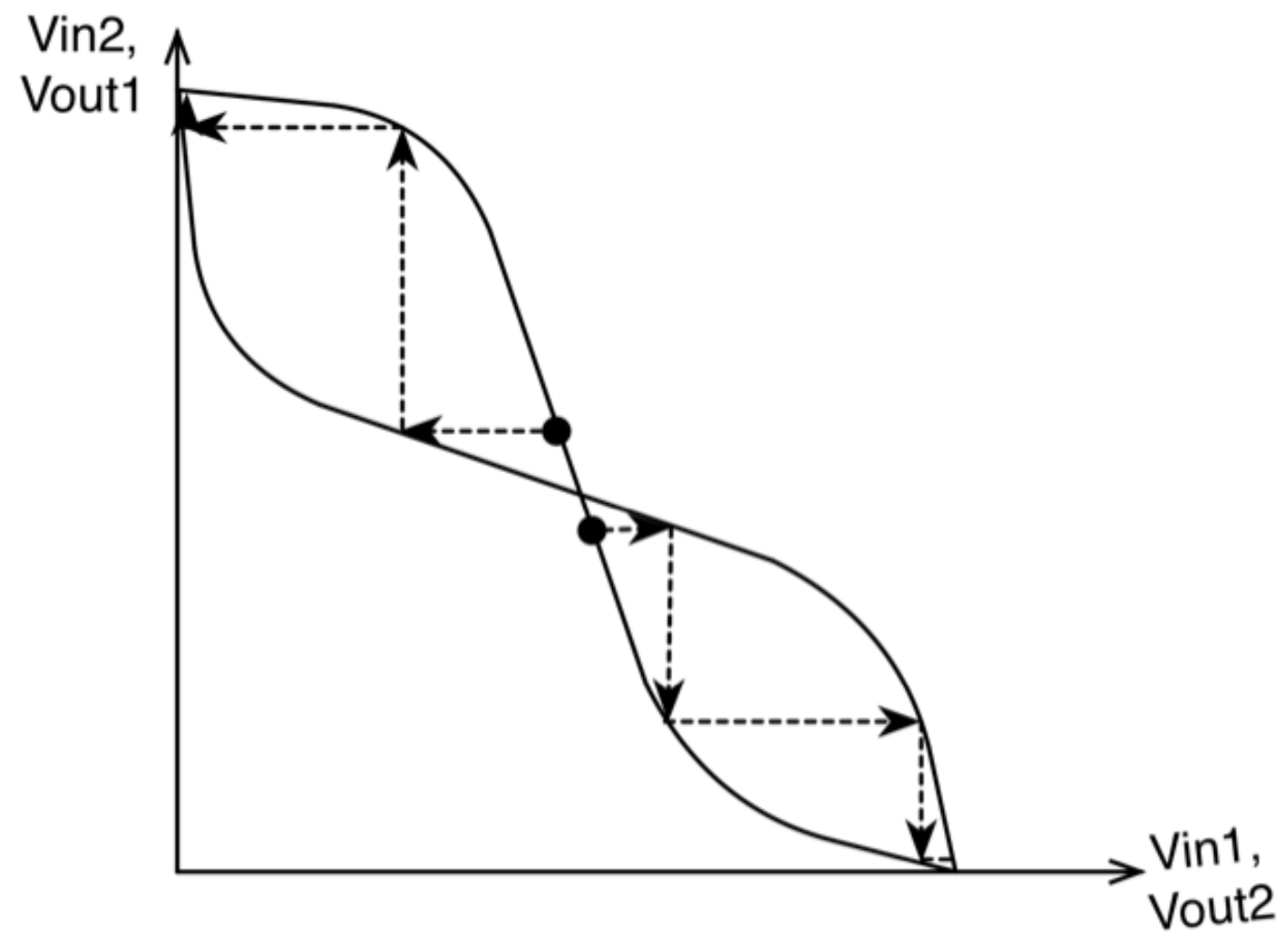
WHAT ELSE DOES VM TELL US?

$$K_n (V_{in} - V_{thn})^2 = K_p (V_{in} - V_{DD} - V_{thp})^2$$

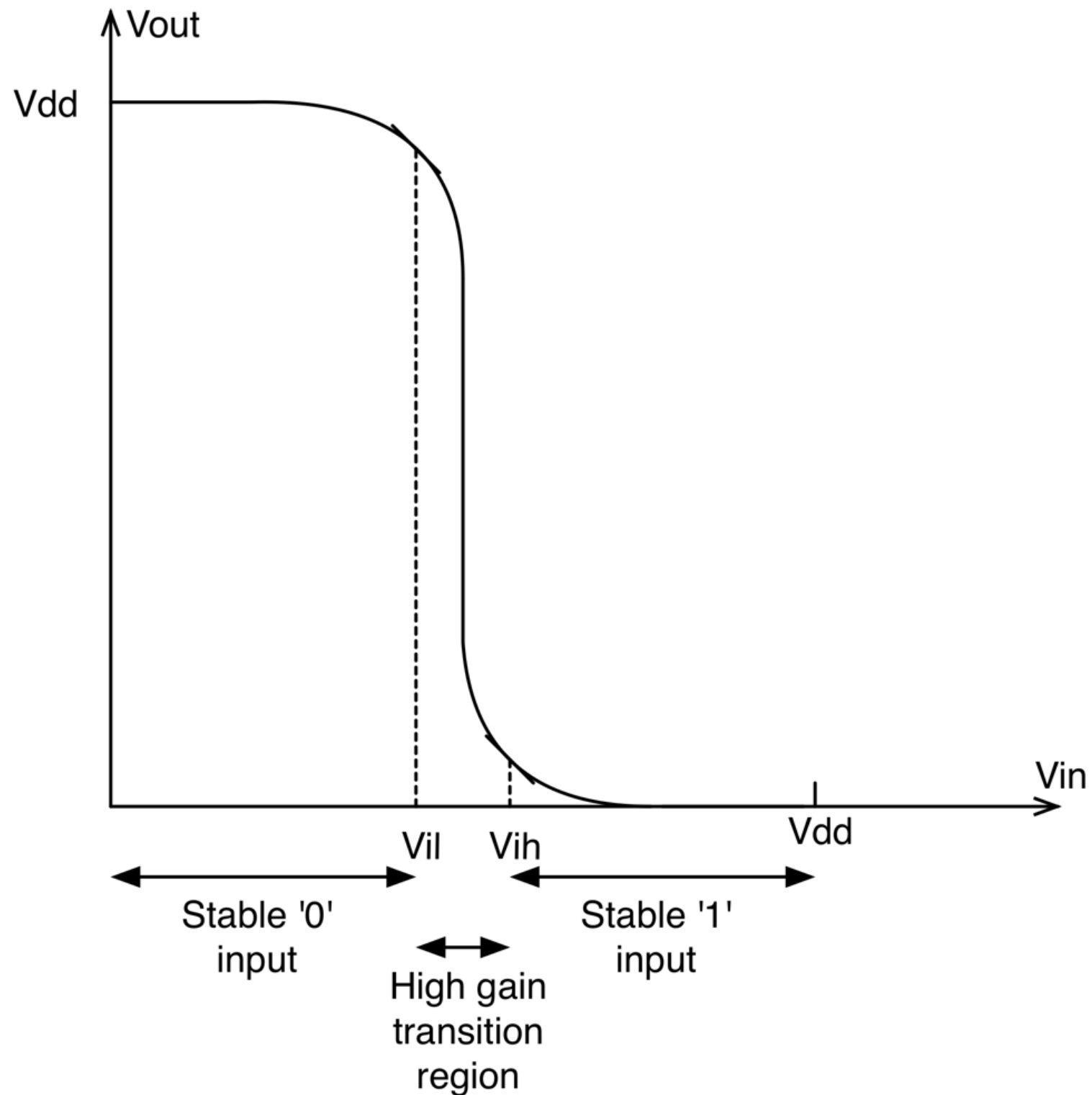
- The equation does not describe a range
- Rather a single point
- The transistors are sat-sat at a single point



USEFULNESS OF HIGH GAIN

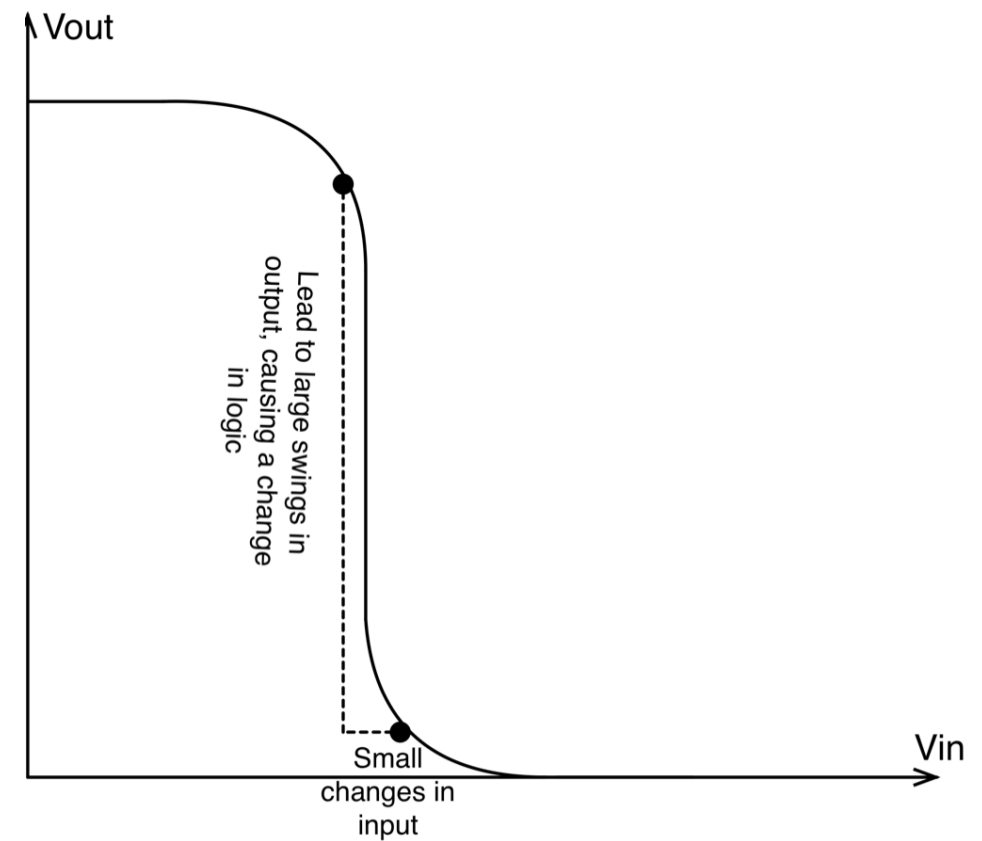


DEFINITION OF NOISE MARGINS



WHY AVOID THE HIGH GAIN REGION?

- Small changes in input lead to wild swings in output
- Sensitivity to noise is very high



OBTAINING VIL

$$I_{Mn} \Big|_{ohmic} = I_{Mp} \Big|_{sat}$$

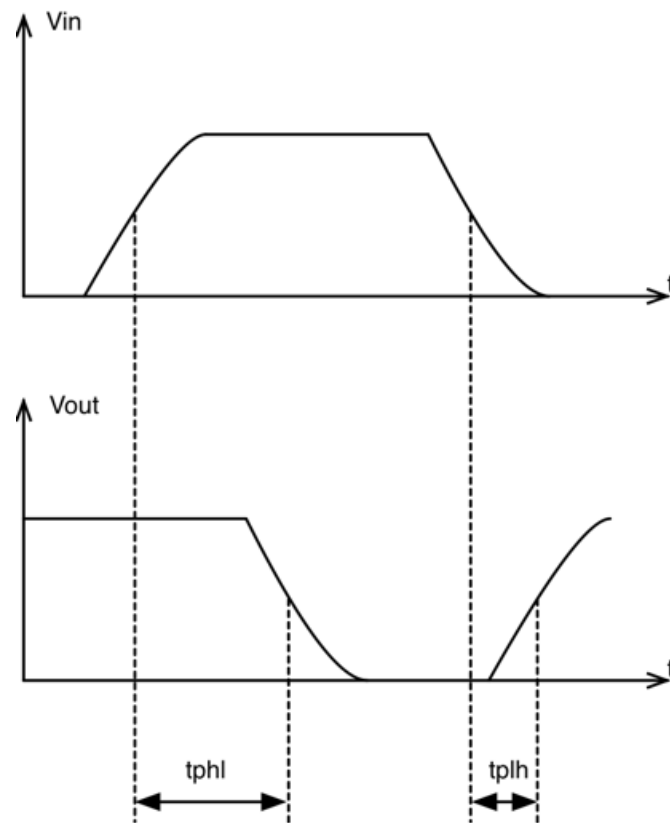
$$2K_n((V_{in} - V_{thn})V_{out} - 0.5V_{out}^2) = K_p(V_{in} - V_{DD} - V_{thp})^2$$

$$V'_{out} = -1$$

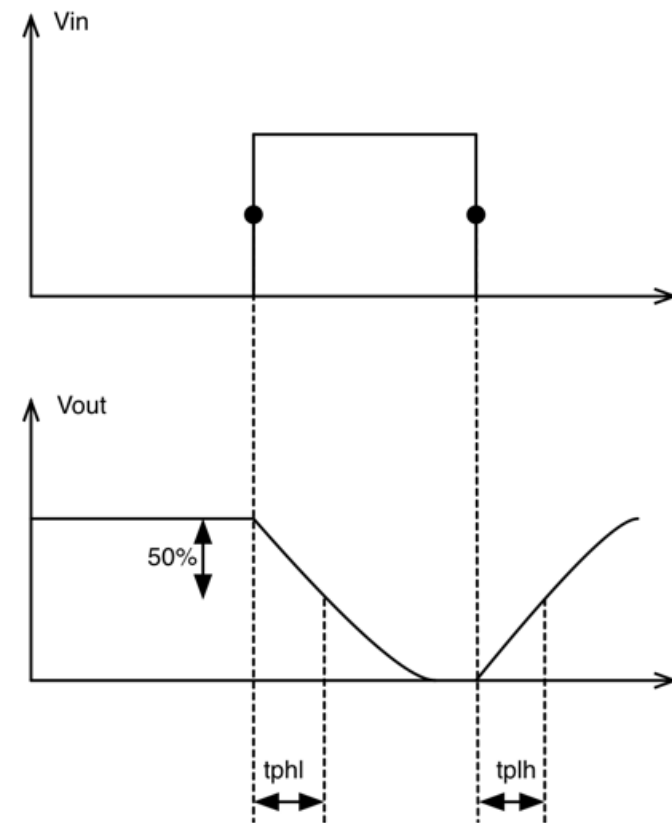
$$2K_n(V_{out} + (V_{in} - V_{thn})V'_{out} - V_{out}V'_{out}) = 2K_p(V_{in} - V_{DD} - V_{thp})$$

$$2K_n(2V_{out} - V_{in} + V_{thn}) = 2K_p(V_{in} - V_{DD} - V_{thp})$$

DEFINITION OF DELAY



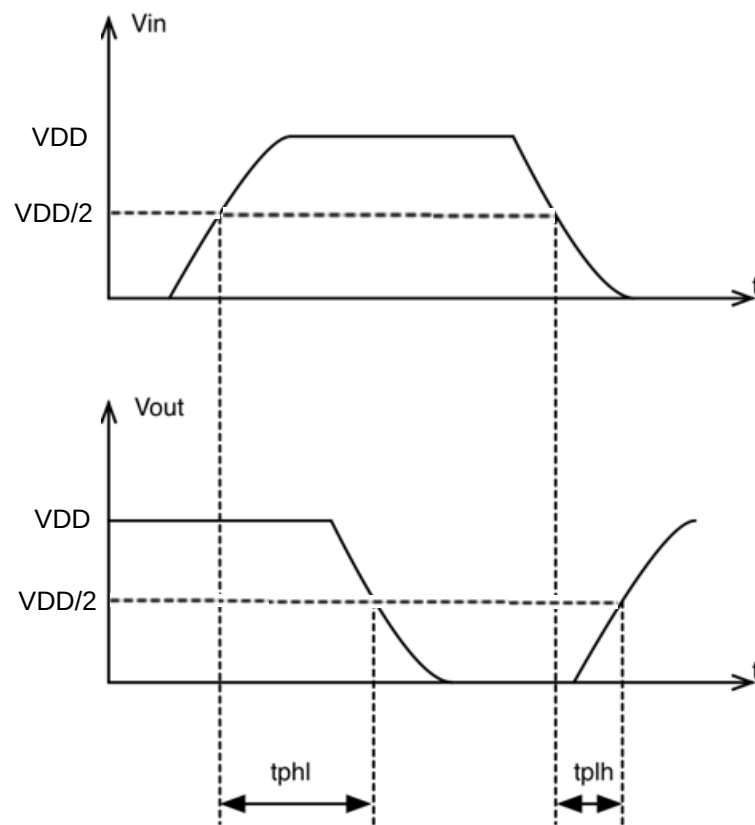
Finite slope input



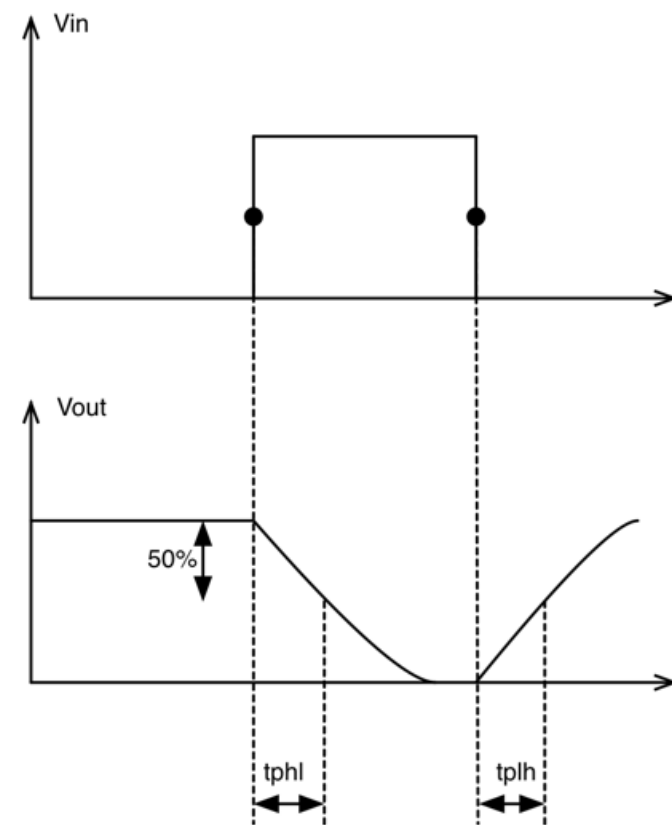
Sharp input

- Distinguish rise and fall times from propagation delays

IDEAL INPUTS



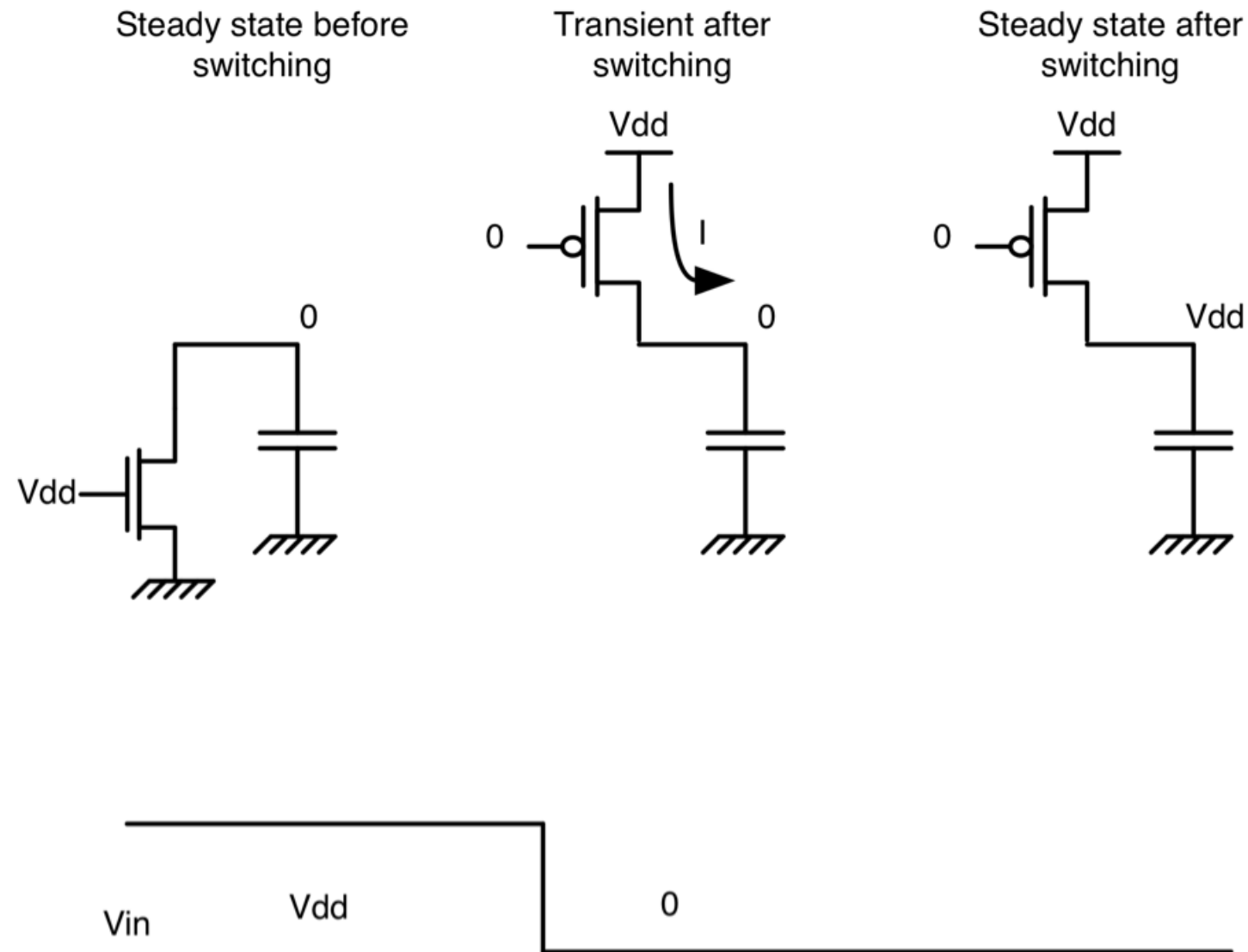
Finite slope input



Sharp input

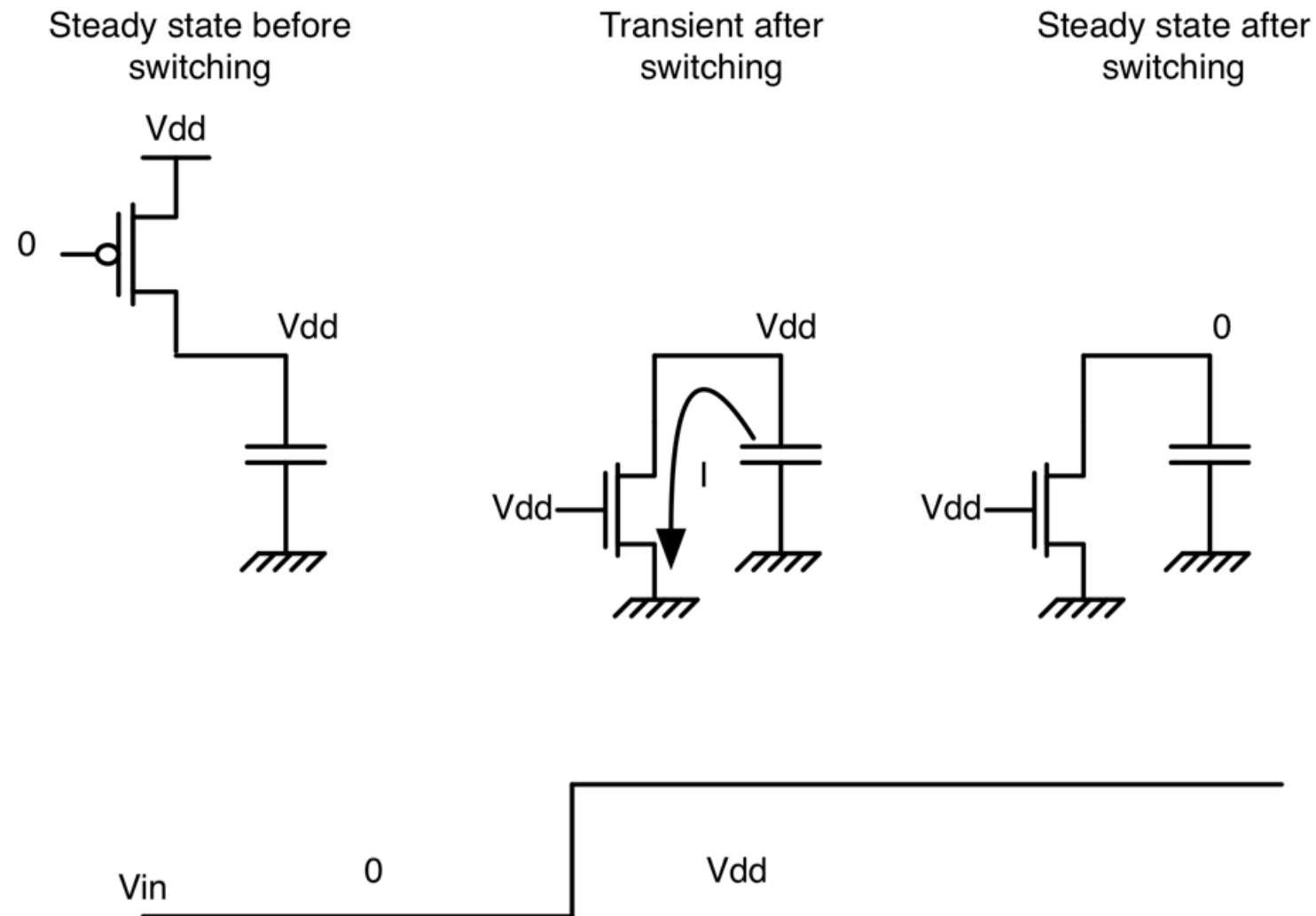
- If the inputs are ideally sharp propagation delays reduce to rise and fall times

CMOS INVERTER INPUT SWITCHING FROM '1' TO '0'



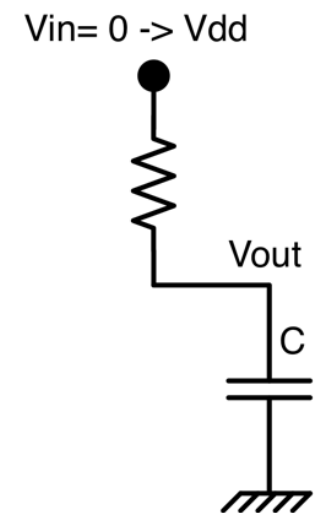
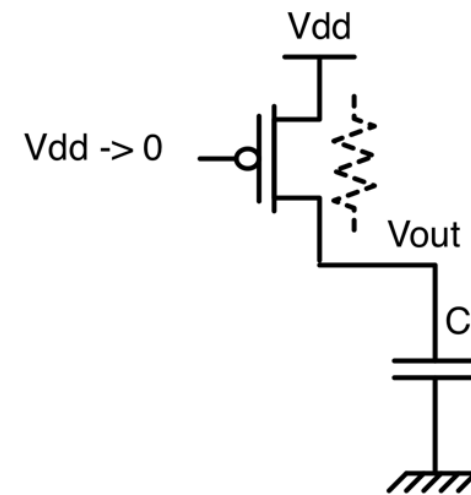
CMOS INVERTER INPUT SWITCHING FROM '0' TO '1'

.....

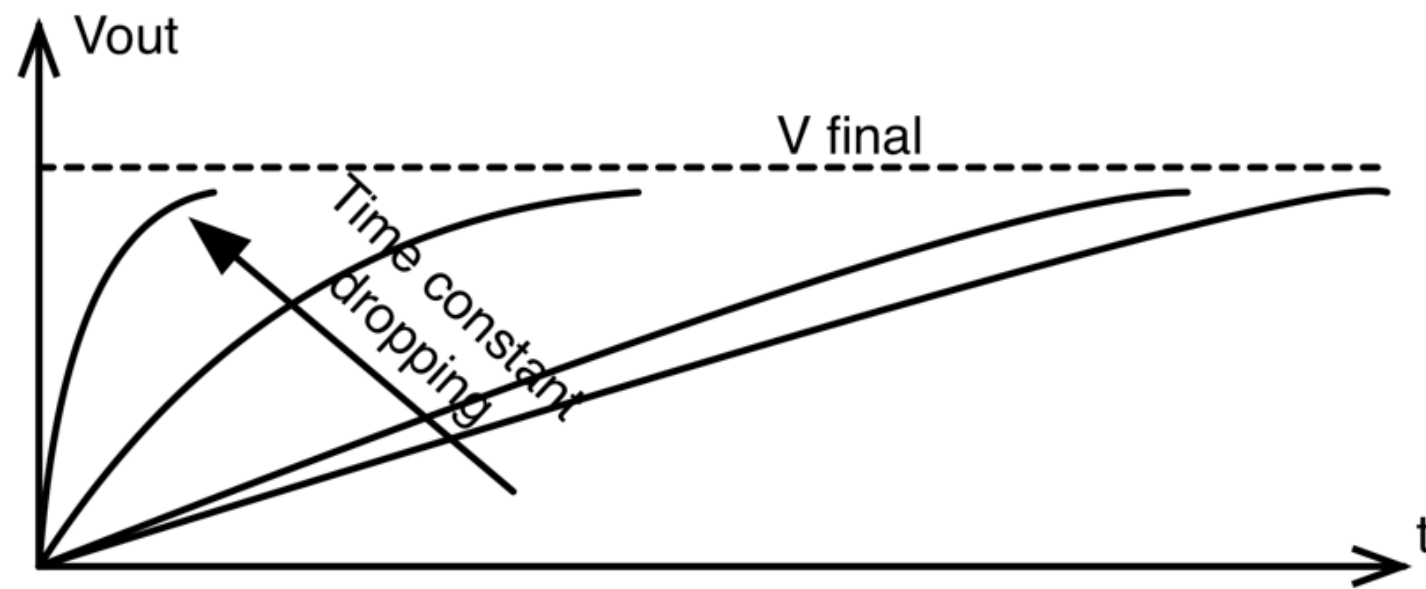


RC EQUIVALENT CIRCUIT

- R is not linear
- We still do not know how to estimate C



STEP RESPONSE OF RC CIRCUIT



STEP RESPONSE OF RC CIRCUIT

$$\frac{V_{in} - V_{out}}{R} = C \frac{dV_{out}}{dt}$$

$$V_{out} = V_{DD}(1 - e^{-t/RC}) = V_{DD}(1 - e^{-t/\tau})$$

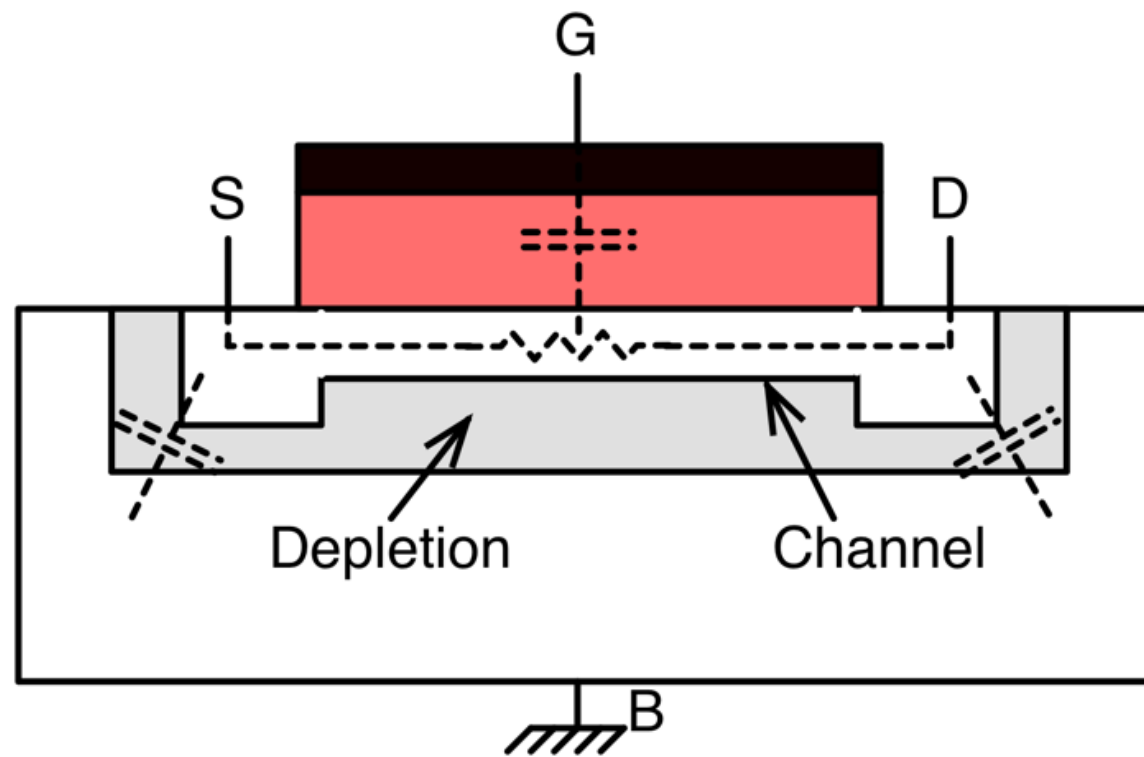
LINEAR APPROXIMATION FOR 50% RISE/FALL

$$t_{phl} = 0.69\tau_n = 0.69R_nC_L$$

$$t_{plh} = 0.69\tau_p = 0.69R_pC_L$$

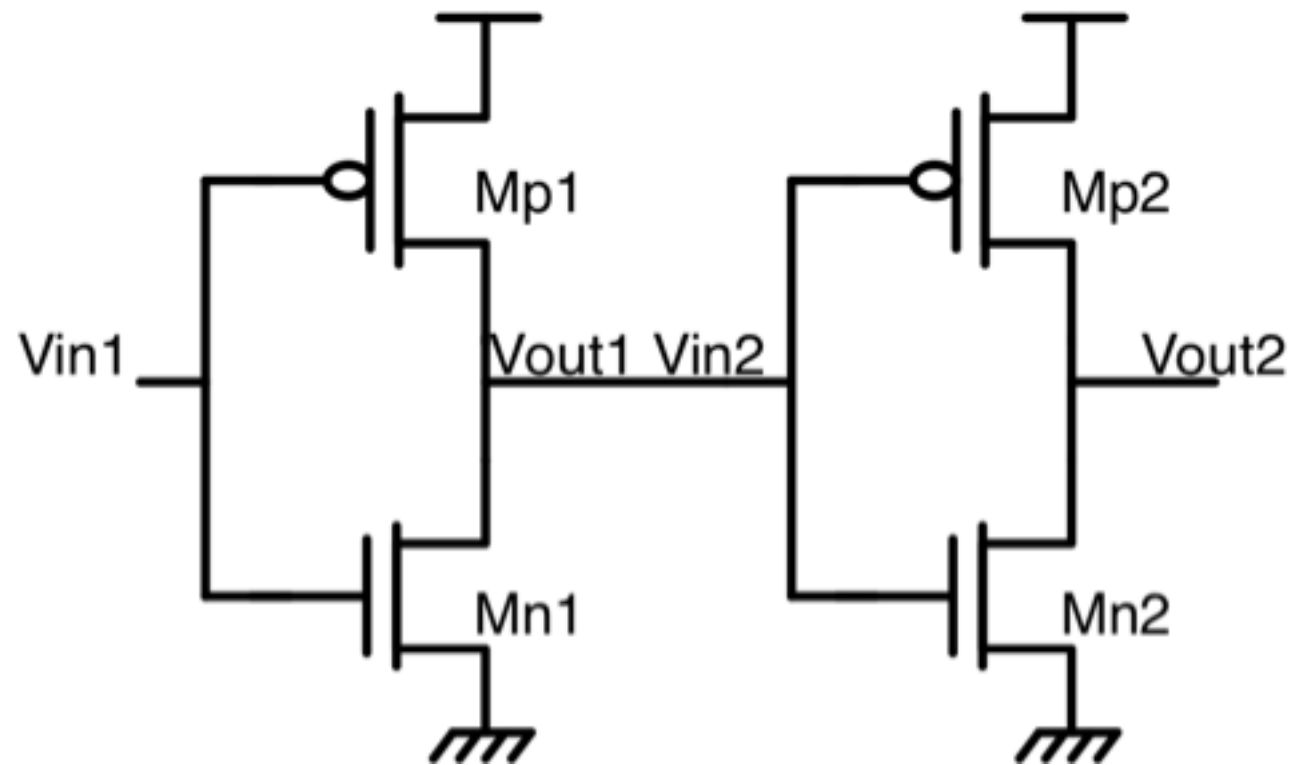
➤ We still have not estimated either R or CL

CAPACITANCES AND RESISTANCE



- Resistance in the MOSFET channel
- Gate capacitance
- Diffusion capacitances from source and drain to body

TYPICAL LOADING SCENARIO



- A CMOS gate loaded by an identical gate
- At V_{out1} , $Mn1$ and $Mp1$ charge/discharge node
- Node capacitance formed by C_{dn1} , C_{dp1} , C_{gn2} , C_{gp2}

THE PROBLEM WITH RESISTANCE

- It is non-linear
- Calculate an average value

$$R_{Mn}(V_{out}) = \frac{V_{out}}{I_{M_n}|_{sat}}$$

$$R_{Mn}(V_{out}) = \frac{V_{out}}{I_{M_n}|_{sat} (1 + \lambda V_{out})}$$

$$R_{eq} = \frac{1}{-V_{DD}/2} \int_{V_{DD}}^{V_{DD}/2} \frac{V}{I_{M_n}|_{sat} (1 + \lambda V)} dV$$

$$R_{eq} = \frac{3}{4} \frac{V_{DD}}{I_{M_n}|_{sat}} \left(1 - \frac{7}{9} \lambda V_{DD}\right)$$

QUESTIONS ABOUT RESISTANCE

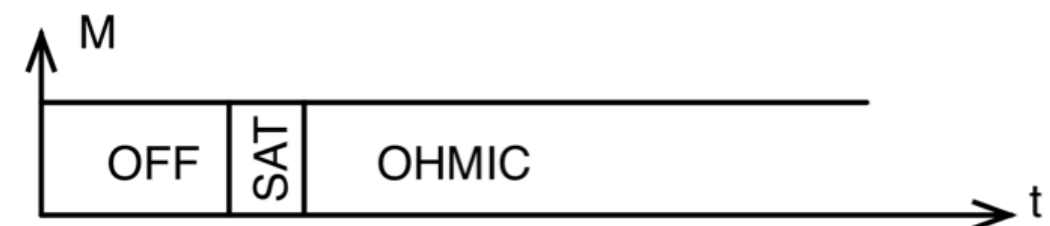
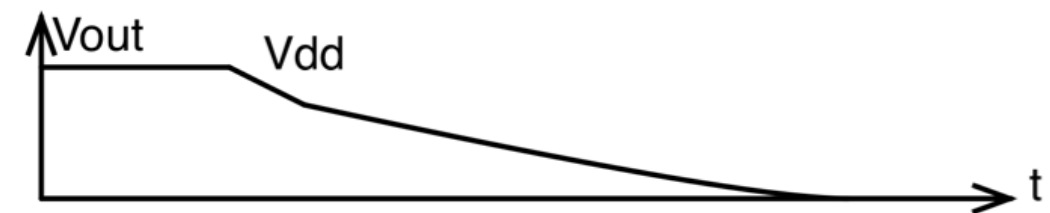
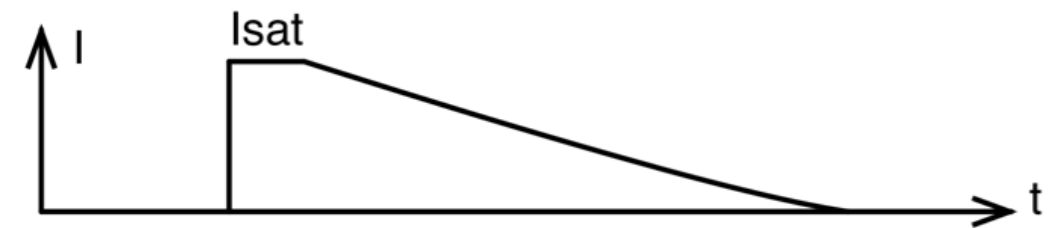
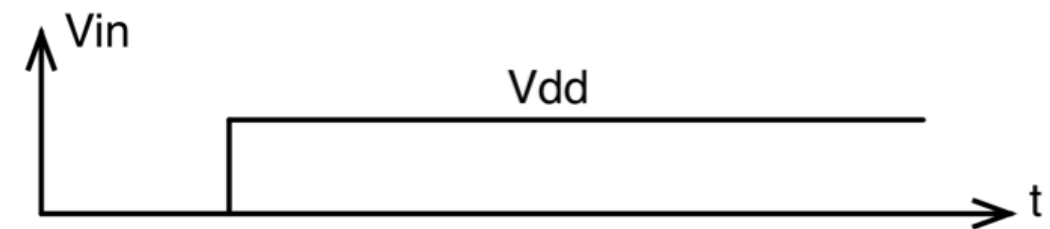
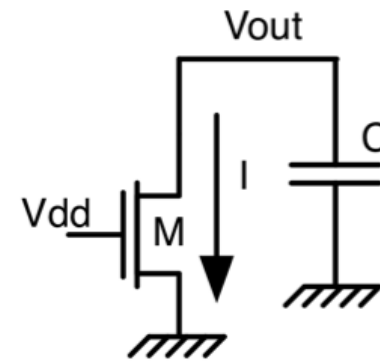
- Why are we using saturation current?
 - Highest available current
 - Responsible for most of the discharge
 - This is more true in short channel transistors

EXAMINING RESISTANCE

$$R_{eq} = \frac{3V_{DD}}{2K_n(V_{gs} - V_{thn})^2} = \frac{3V_{DD}}{2\mu_n C_{ox}(W/L)_n(V_{DD} - V_{thn})^2}$$

- R is inversely proportional to W
- R is directly proportional to L
- An NMOS transistor has half the resistance of a similarly sized PMOS transistor

BEHAVIOR DURING DISCHARGE



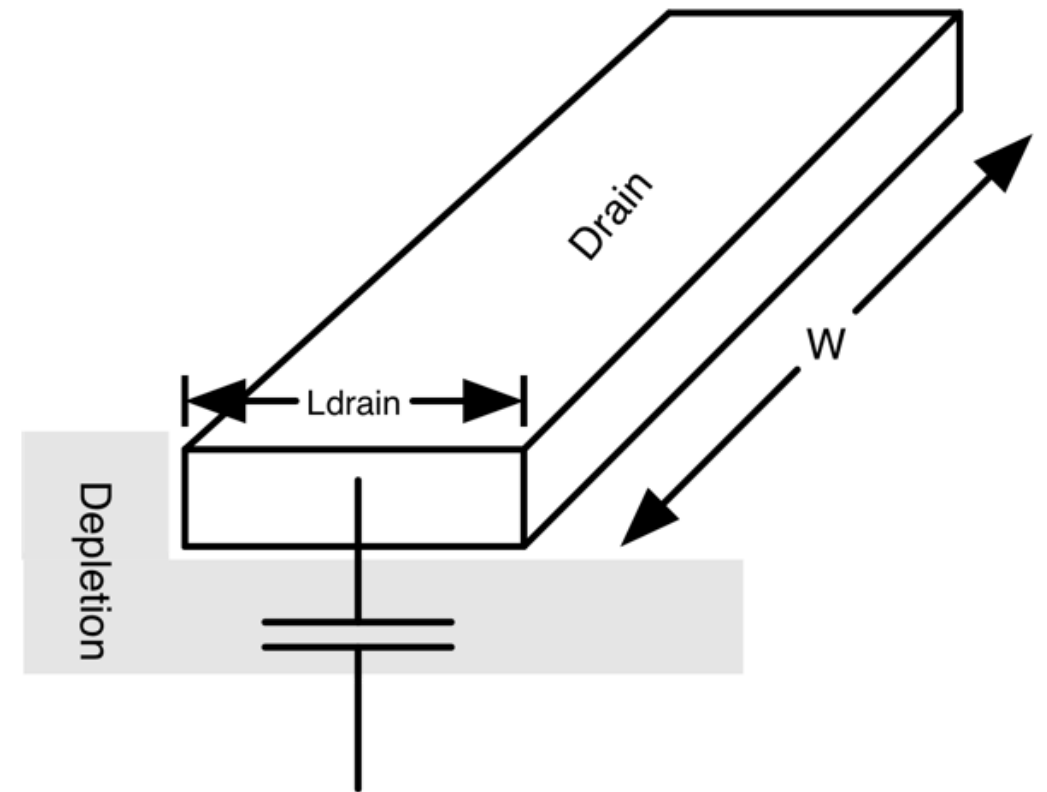
GATE CAPACITANCE

$$C_{gate} = C_{ox}WL$$

- There are two major simplifications here:
 - Assume all gate capacitance appears to the source
 - Assume the gate capacitance appears to ground

JUNCTION CAPACITANCE

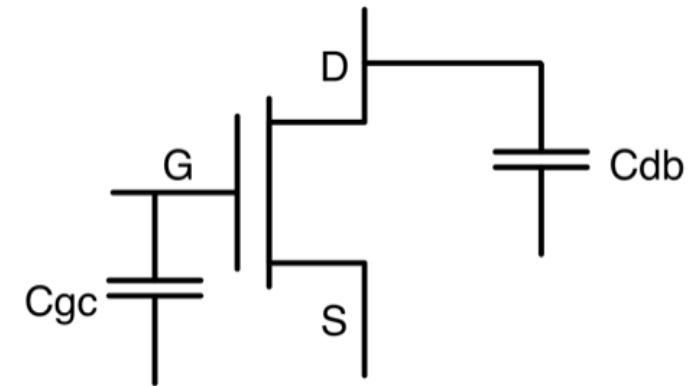
- Directly proportional to W
- No relation to L
- There is a lot of information in C_j



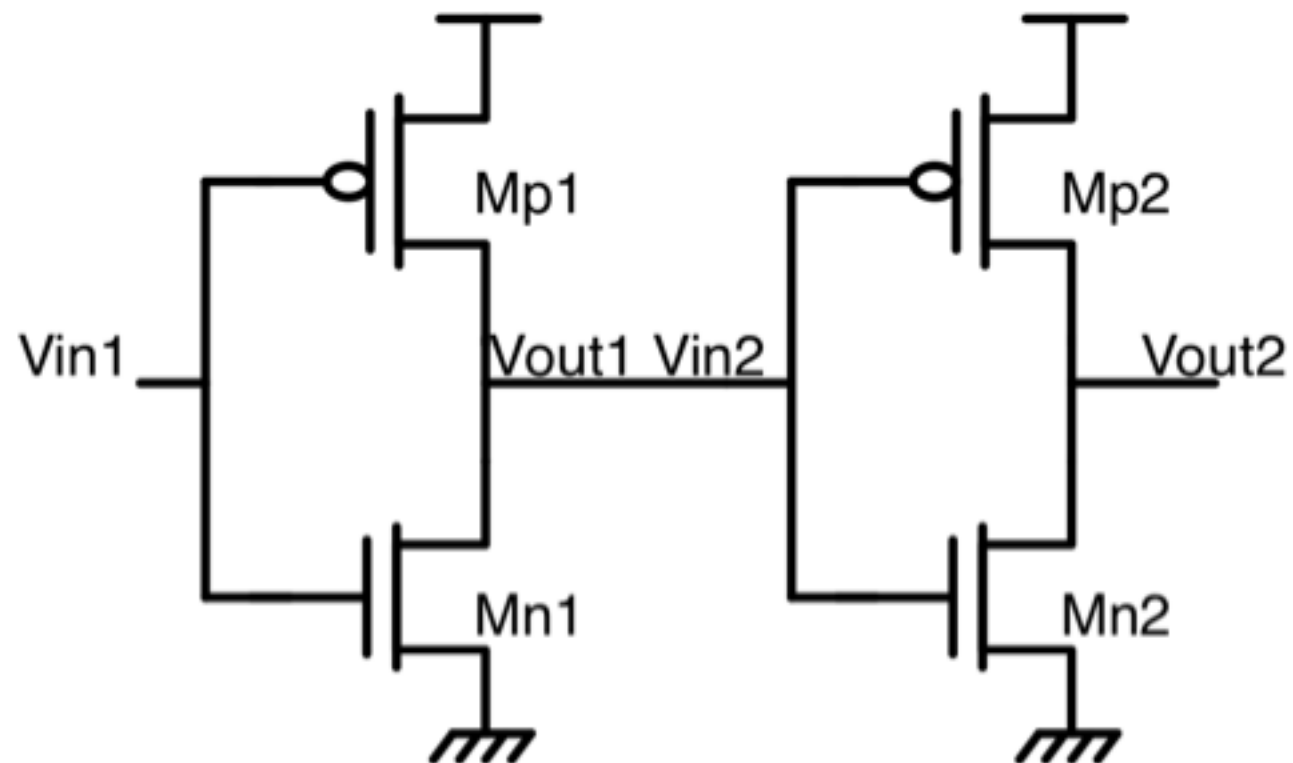
$$C_{drain} = C_j W L_{drain}$$

SCHEMATIC MODEL

- Gate capacitance should be divided between source and drain
 - Assume appears to ground
- C_{db} appears to ground
- Source capacitance exists
 - Affects only internal nodes

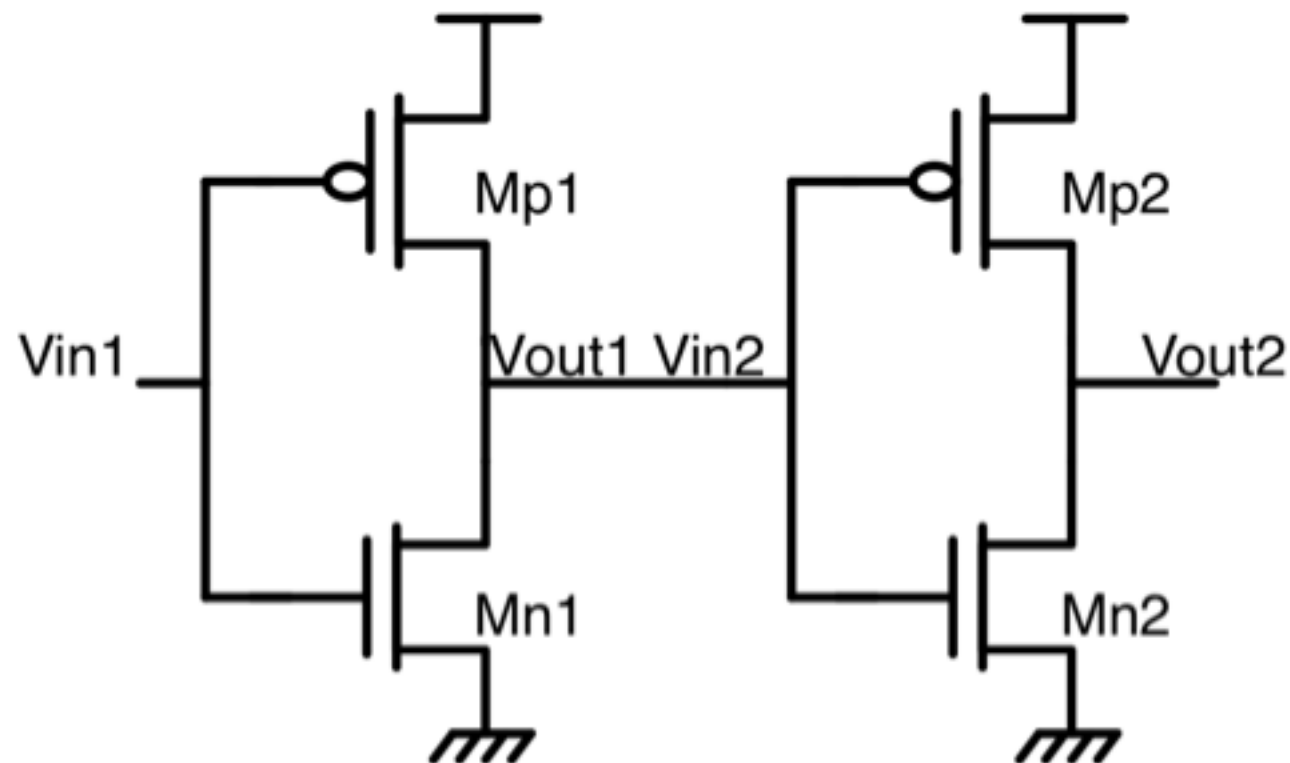


WHAT IS THE TOTAL LOADING AT VOUT1



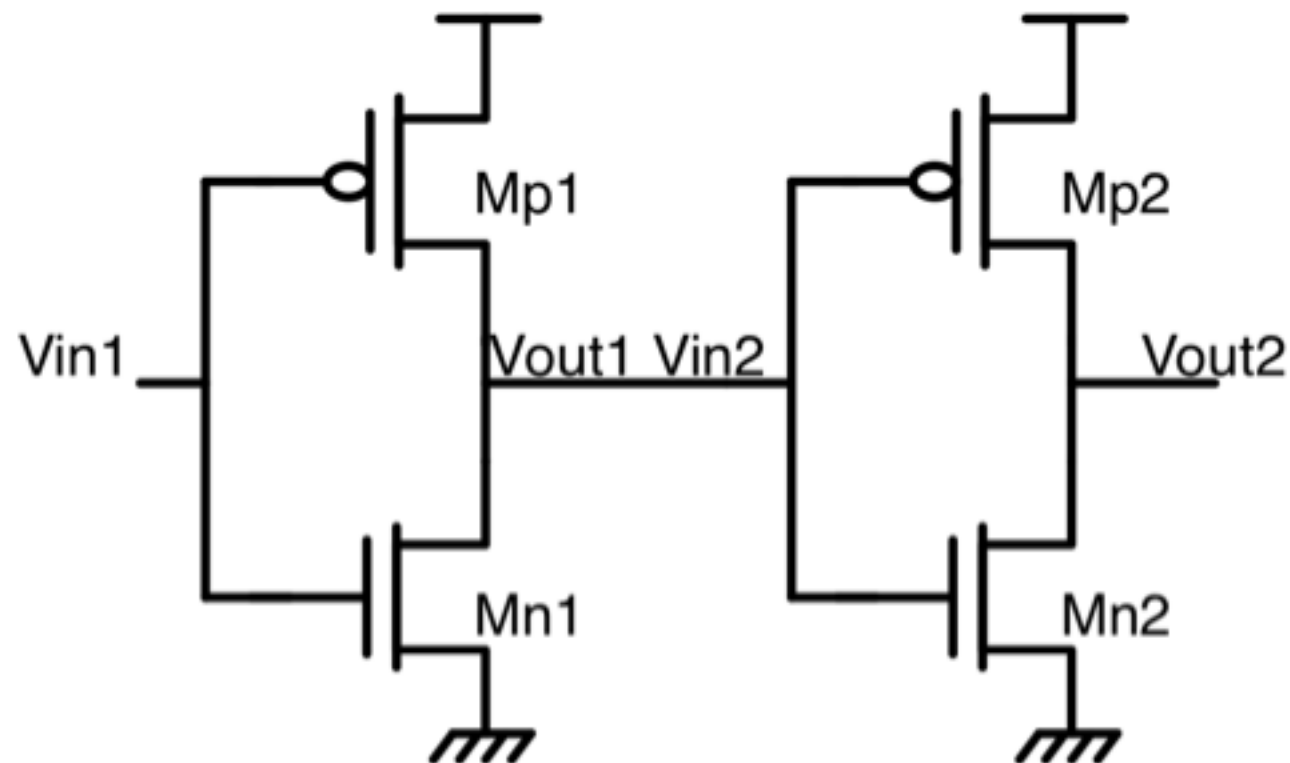
$$C_{out1} = C_{gn2} + C_{gp2} + C_{dn1} + C_{dp1}$$

SELF AND EXTERNAL LOADING



- Drain capacitance always comes from current stage
- Gate capacitance always comes from next stage

WHICH RESISTANCE TO USE



$$t_{phl} = 0.69R_{n1}C_{out1}$$

$$t_{plh} = 0.69R_{p1}C_{out1}$$

IMPACT OF W AND L

- There is no advantage to using $L > L_{min}$
- W increasing leads to conflicting effects

	C _{gate}	C _{drain}	R
L increases	Increases	N/A	Increases
W increases	Increases	Increases	Decreases

THE UNIT TRANSISTOR

$$C_{dn0}, C_{dp0}, C_{gn0}, C_{gp0}, R_{n0}, R_{p0}$$

$$R_{p0} = 2R_{n0}$$

$$C_g = C_{g0} * (W/L)$$

$$R_n = \frac{R_{n0}}{(W/L)}$$

➤ Transistor with $W=L=L_{min}$

WHY WE CAN USE W/L EVEN FOR CDB

$$C_1 = C_0 W_1$$

$$C_2 = C_0 W_2$$

$$\frac{C_1}{C_2} = \frac{C_0 W_1}{C_0 W_2}$$

$$\frac{C_1}{C_2} = \frac{(W_1/L_{\min})}{(W_2/L_{\min})}$$

RECAP

- Given capacitances and resistance for unit transistors:
 - Find actual resistances and capacitances in proportion to W/L
 - Calculate CL
 - Calculate time constant
 - Calculate propagation delays

POWER CANNOT BE NULL

- While static current (and power) is null, total power cannot be null
- All commercial electronics are CMOS
- They all run out of juice
- There must be power dissipation

WHERE IS POWER BEING DISSIPATED

- We already know where
- The question is when
- Whenever current flows in MOSFET, power will be dissipated
- Power will dissipate when switching state
- Dynamic power

CONSERVATION OF ENERGY

- Conservation of energy states
 - Power drawn from supply = power dissipated in resistors
- Valid only when the circuit is in steady state
- In transience and in the presence of reactive elements:
 - Power drawn from supply = power dissipated in resistors
+ power stored in reactive elements

SUPPLIED ENERGY IN LH TRANSITION

$$E_{drawn} = \int I_{drawn} V_{DD} dt$$

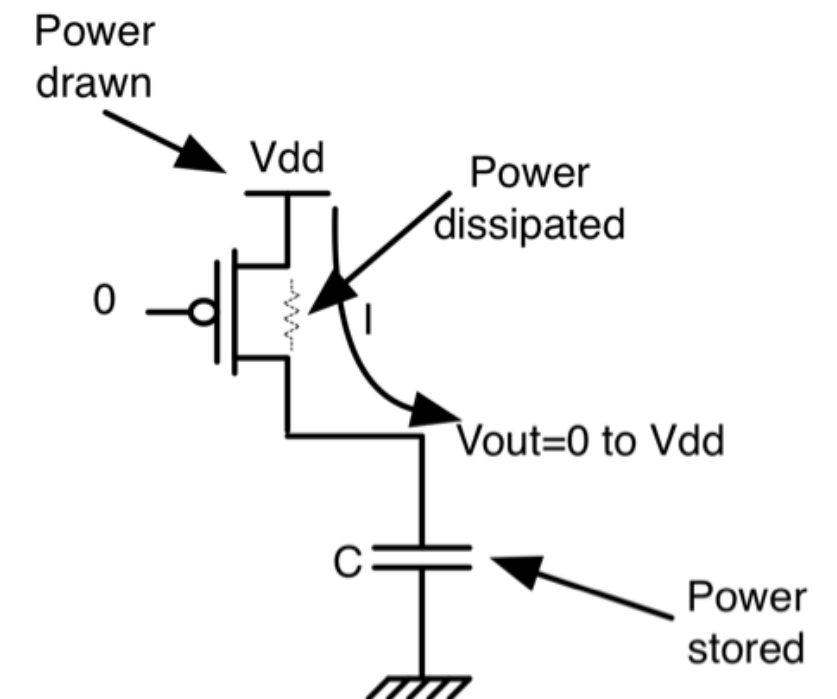
$$I_{drawn} = I_{Mp} = I_C = C \frac{dV_{out}}{dt}$$

$$E_{drawn} = CV_{DD} \int \frac{dV_{out}}{dt} dt$$

$$E_{drawn} = CV_{DD} \int_{t=0}^{t=\infty} \frac{dV_{out}}{dt} dt$$

$$E_{drawn} = CV_{DD} \int_{V_{out}=0}^{V_{out}=V_{DD}} dV_{out}$$

$$E_{drawn} = CV_{DD}^2$$



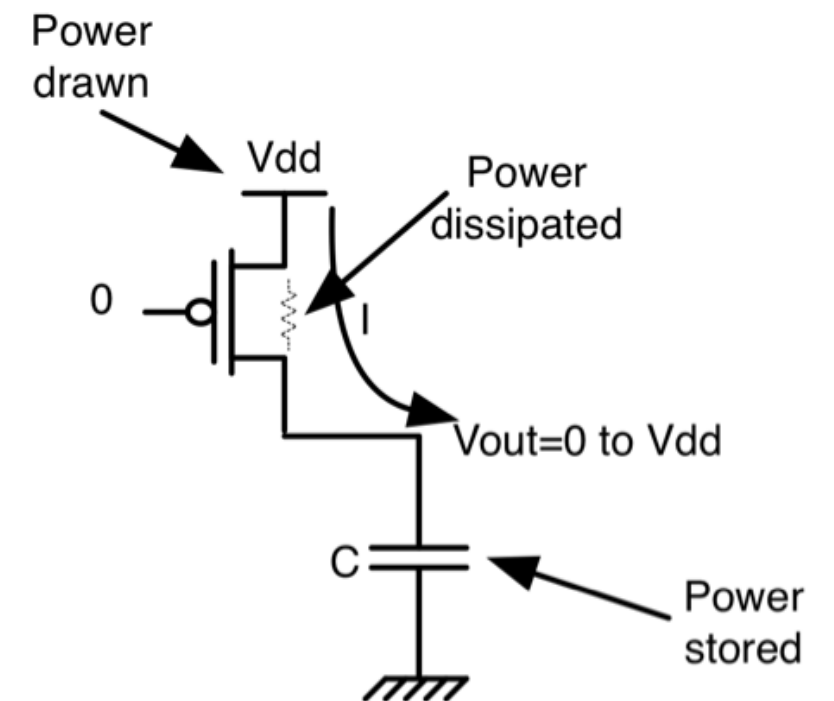
STORED ENERGY IN LH TRANSITION

$$E_{stored} = \int_{t=0}^{t=\infty} I_C V_{out} dt$$

$$E_{stored} = \int_{t=0}^{t=\infty} C \frac{dV_{out}}{dt} V_{out} dt$$

$$E_{stored} = C \int_{V_{out}=0}^{V_{out}=V_{DD}} V_{out} dV_{out}$$

$$E_{stored} = C \frac{V_{DD}^2}{2}$$



DISSIPATED ENERGY DURING LH TRANSITION

$$E_{drawn} = E_{stored} + E_{PMOS}$$

$$CV_{DD}^2 = C \frac{V_{DD}^2}{2} + E_{PMOS}$$

$$E_{PMOS} = C \frac{V_{DD}^2}{2}$$

NOTES ON ENERGY DISSIPATION

- Where is this energy dissipated?
 - In the PMOS channel
- Curious how resistance value does not factor in energy
- Why not calculate dissipated energy directly?
 - Current in the MOSFET is non-linear, will be harder to integrate

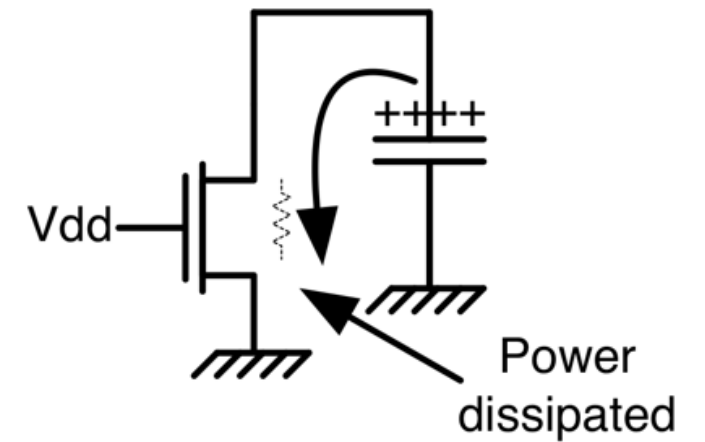
ENERGY DISSIPATED DURING HL TRANSITION

$$E_{C_initial} = E_{C_final} + E_{NMOS}$$

$$E_{C_final} = 0$$

$$E_{NMOS} = E_{C_initial}$$

$$E_{NMOS} = E_{C_initial} = C \frac{V_{DD}^2}{2}$$



TOTAL POWER DISSIPATION FOR A CLOCK INPUT

$$P = (E_{NMOS} + E_{PMOS})/T$$

$$P = CV_{DD}^2/T$$

$$P = CV_{DD}^2 f$$

$$P = \alpha CV_{DD}^2 f$$

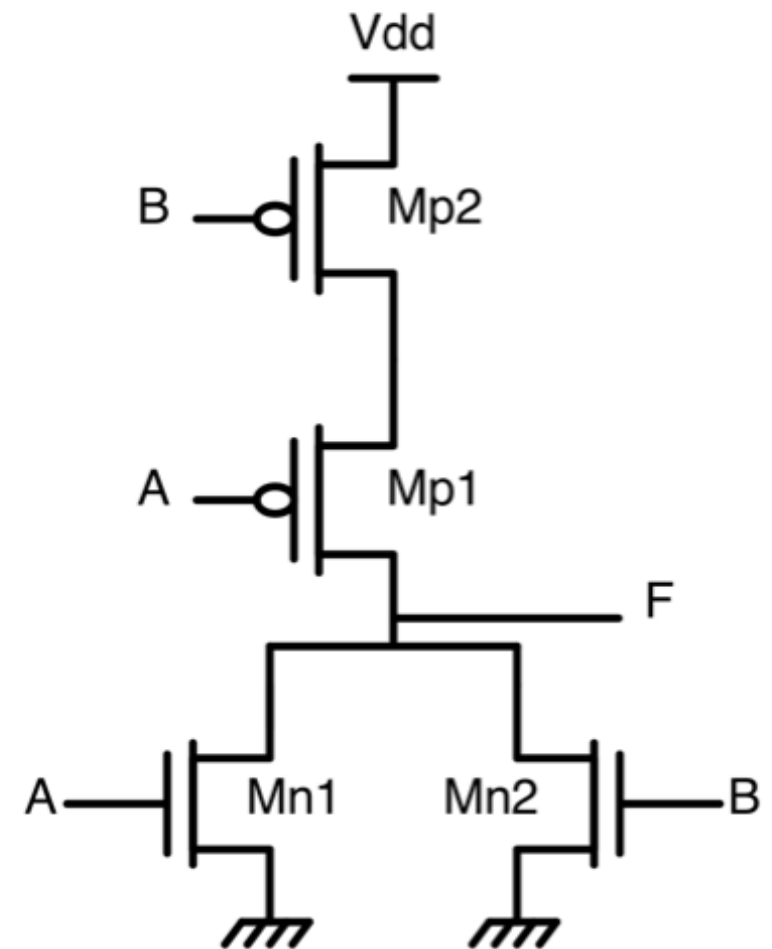
ROLE OF PDN AND PUN

.....

	PDN	PUN
Transistor types	NMOS	PMOS
Transistor count	N	N
Part of truth table covered	0' outputs	1' outputs
Role in delay	R in tpHL	R in tpLH

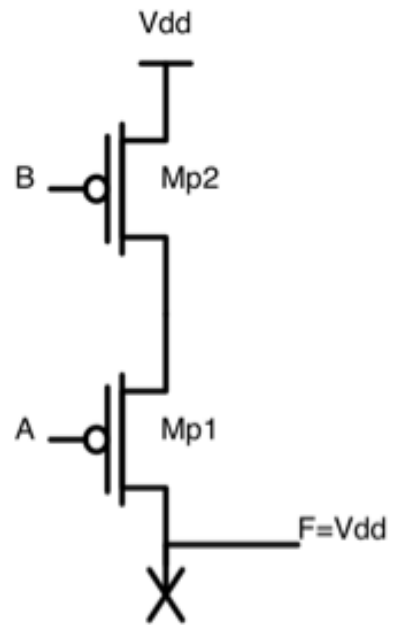
CMOS NOR GATE

- Note complementary connections

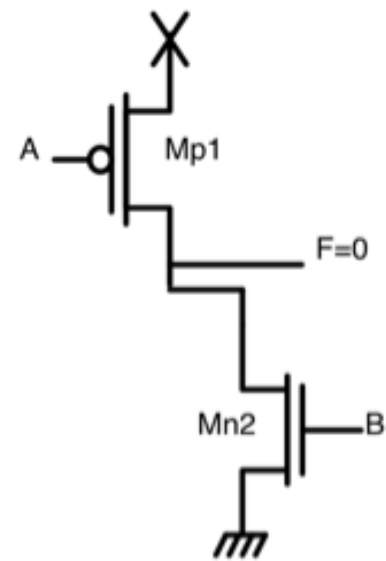


FOUR CASES OF CMOS NOR

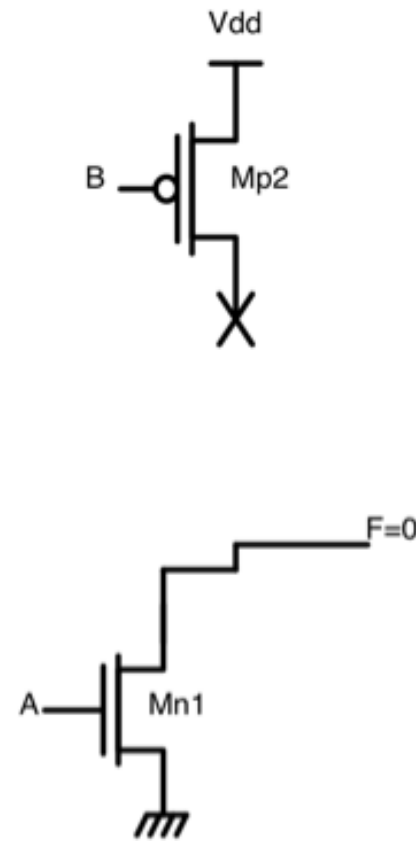
- Recall parallel and series equivalence



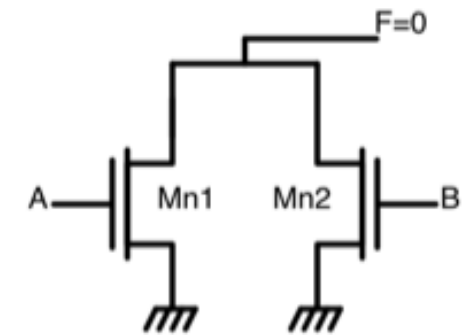
$AB = '00'$



$AB = '01'$



$AB = '10'$



$AB = '11'$

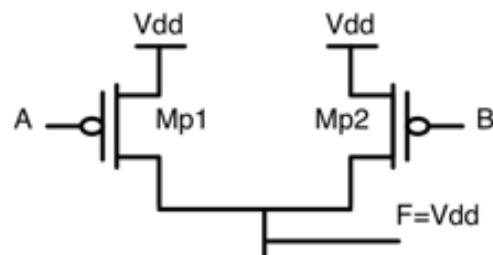
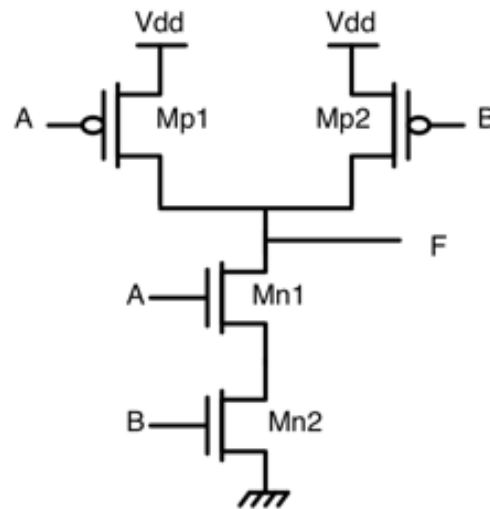
ELECTRIC TRUTH TABLE FOR CMOS NOR

- Note that in all cases zero current flows
 - This is a defining feature of CMOS
 - This is necessary for rail-to-rail logic

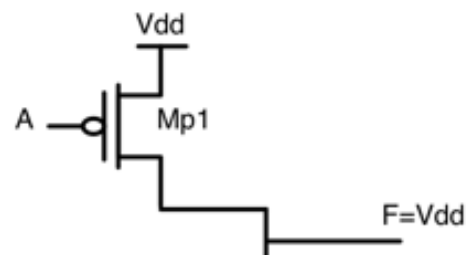
AB	Mn1	Mn2	Mp1	Mp2	Current	Logic out	Vout
00'	OFF	OFF	Ohmic	Ohmic	Zero	1'	Vdd
01'	OFF	Ohmic	Ohmic	OFF	Zero	0'	0
10'	Ohmic	OFF	OFF	Ohmic	Zero	0'	0
11'	Ohmic	Ohmic	OFF	OFF	Zero	0'	0

THE FOUR CASES OF CMOS NAND

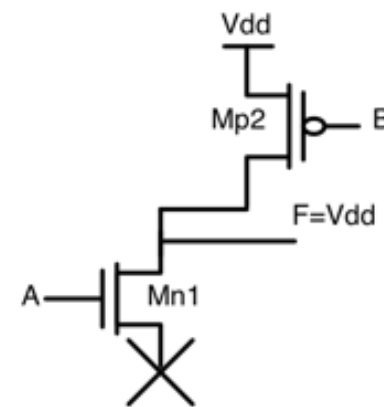
- Use series and parallel reduction to always reach topology similar to inverter



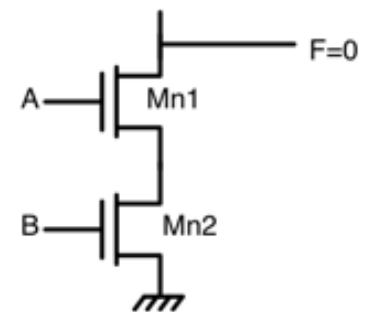
AB='00'



AB='01'



AB='10'



AB='11'

ELECTRIC TRUTH TABLE FOR CMOS NAND

- Still no steady-state current for any input
- All transistors in the steady-state must be either ohmic or cutoff

AB	Mn1	Mn2	Mp1	Mp2	Current	Logic out	Vout
00'	OFF	OFF	Ohmic	Ohmic	Zero	1'	Vdd
01'	OFF	Ohmic	Ohmic	OFF	Zero	1'	Vdd
10'	Ohmic	OFF	OFF	Ohmic	Zero	1'	Vdd
11'	Ohmic	Ohmic	OFF	OFF	Zero	0'	0

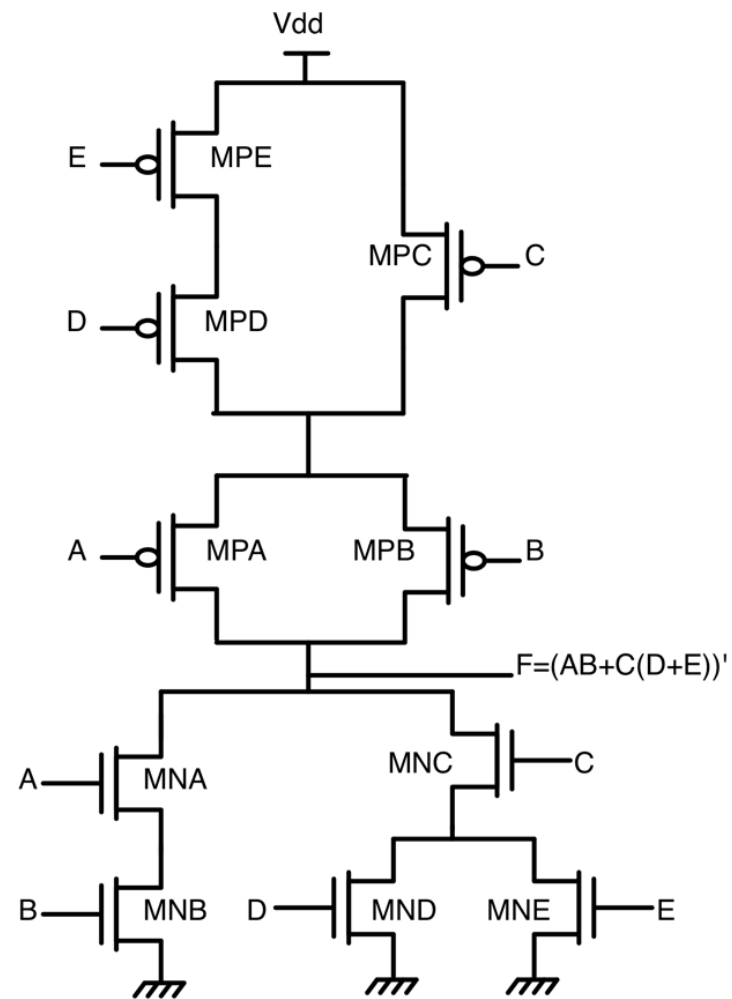
STEPS TO SYNTHESIZE A LOGIC GATE

- PDN as in ratioed logic
 - Obtain the expression of F'
 - Use DeMorgan if necessary
 - Implement PDN
- Implement PUN using complementary connections
 - Check PUN using expression of F

EXAMPLE

- Implement $F = (AB + C(D + E))'$

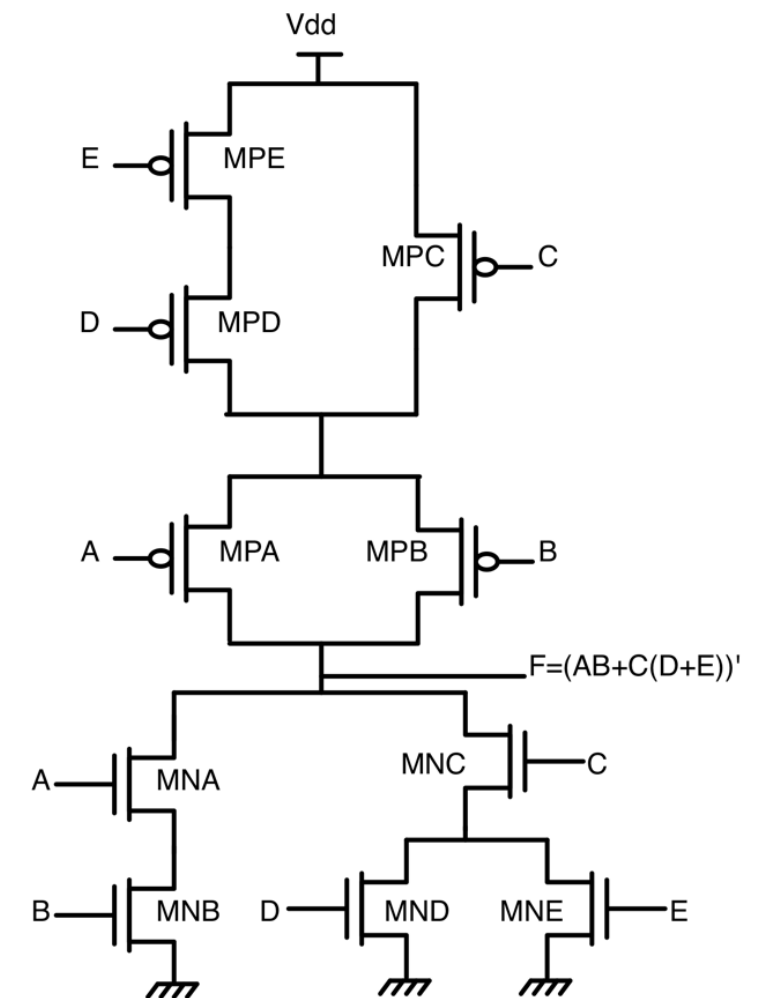
EXAMPLE



EXAMPLE

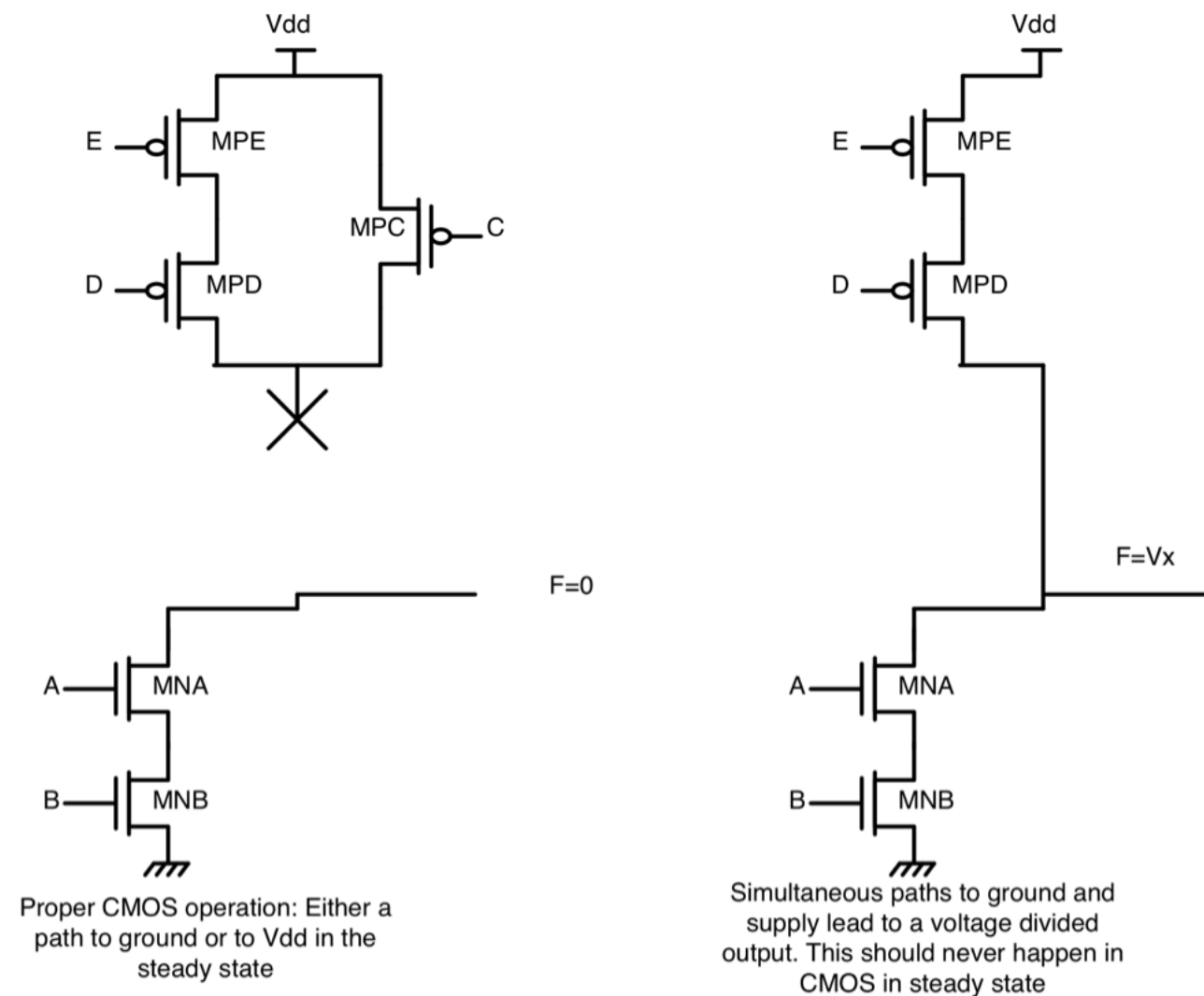
- Never a path from supply to ground
- If paths exist in PDN none can exist in PUN

ABCDE	Number of paths	Number of path	Vout	MNA,MNB,MNC,..	MPA,MPB,MPC,..
00000'	0	2	Vdd	All OFF	All Ohmic
11000'	1	0	0	Ohmic, Ohmic, OFF, OFF, OFF	OFF, OFF, Ohmic, Ohmic, Ohmic
10110'	1	0	0	Ohmic, OFF, Ohmic, Ohmic, OFF	OFF, Ohmic, OFF, OFF, Ohmic
11110'	2	0	0	Ohmic, Ohmic, Ohmic, Ohmic, OFF	OFF, OFF, OFF, OFF, Ohmic
11111'	3	0	0	All Ohmic	All OFF
00011'	0	2	Vdd	OFF, OFF, OFF, Ohmic, Ohmic	Ohmic, Ohmic, Ohmic, OFF, OFF
01011'	0	1	Vdd	OFF, Ohmic, OFF, Ohmic, Ohmic	Ohmic, OFF, Ohmic, OFF, OFF
10001'	0	1	Vdd	Ohmic, OFF, OFF, OFF, Ohmic	OFF, Ohmic, Ohmic, Ohmic, OFF
00001'	0	2	Vdd	OFF, OFF, OFF, OFF, Ohmic	Ohmic, Ohmic, Ohmic, Ohmic, OFF



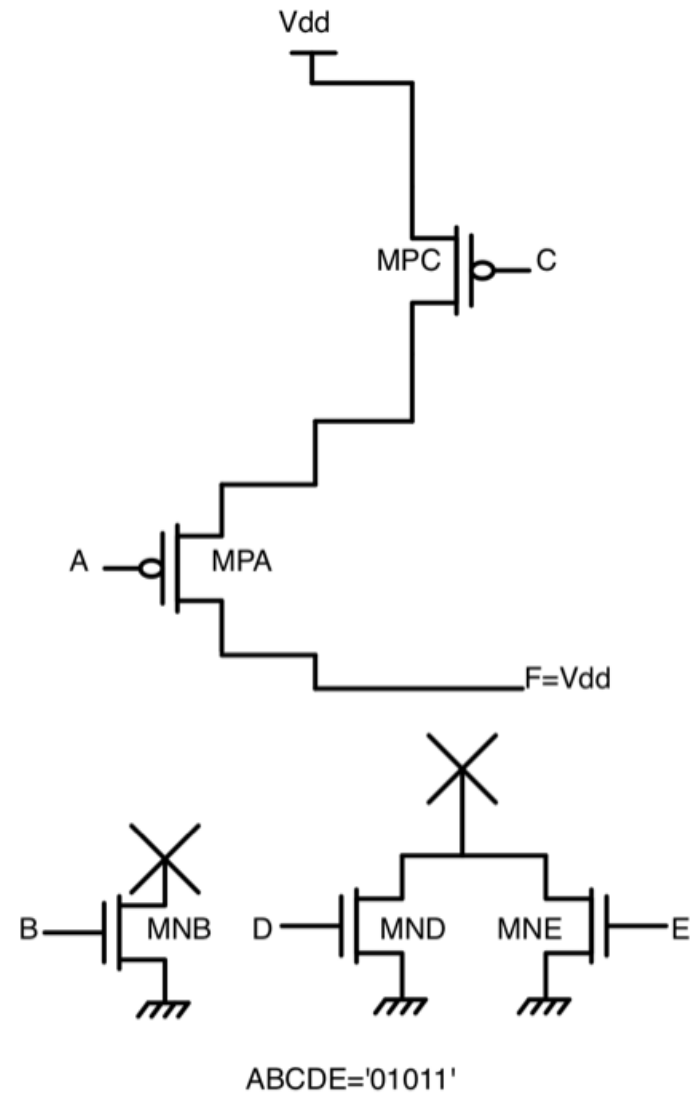
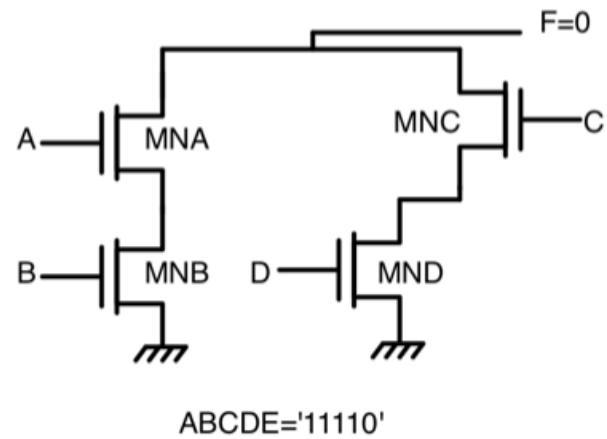
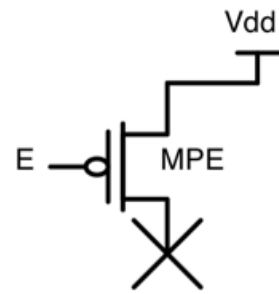
LACK OF STATIC CURRENT IN CMOS

- Static current means a path from supply to ground
- Voltage divider leads to none-rail output



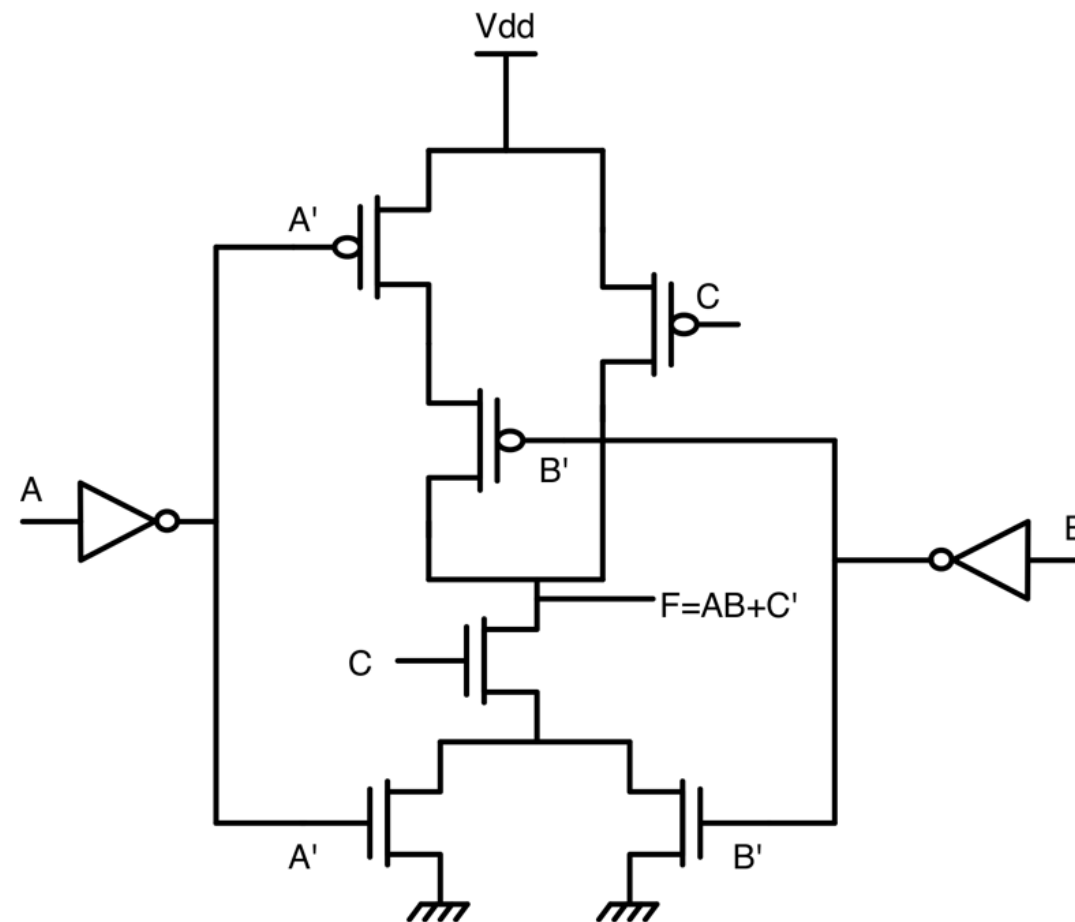
REASONING BEHIND COMPLEMENTARY CONNECTIONS

.....

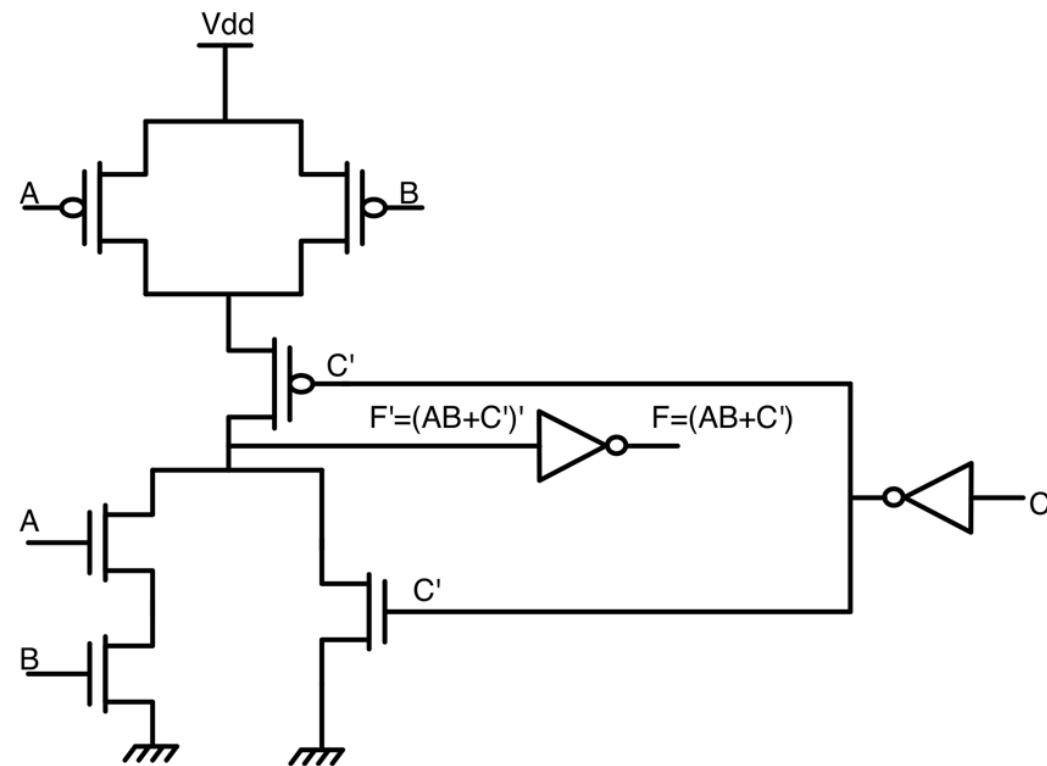


BE WARY OF HIDDEN INVERTERS

.....



ALTERNATIVE IMPLEMENTATION



WHICH IMPLEMENTATION IS “BEST”?

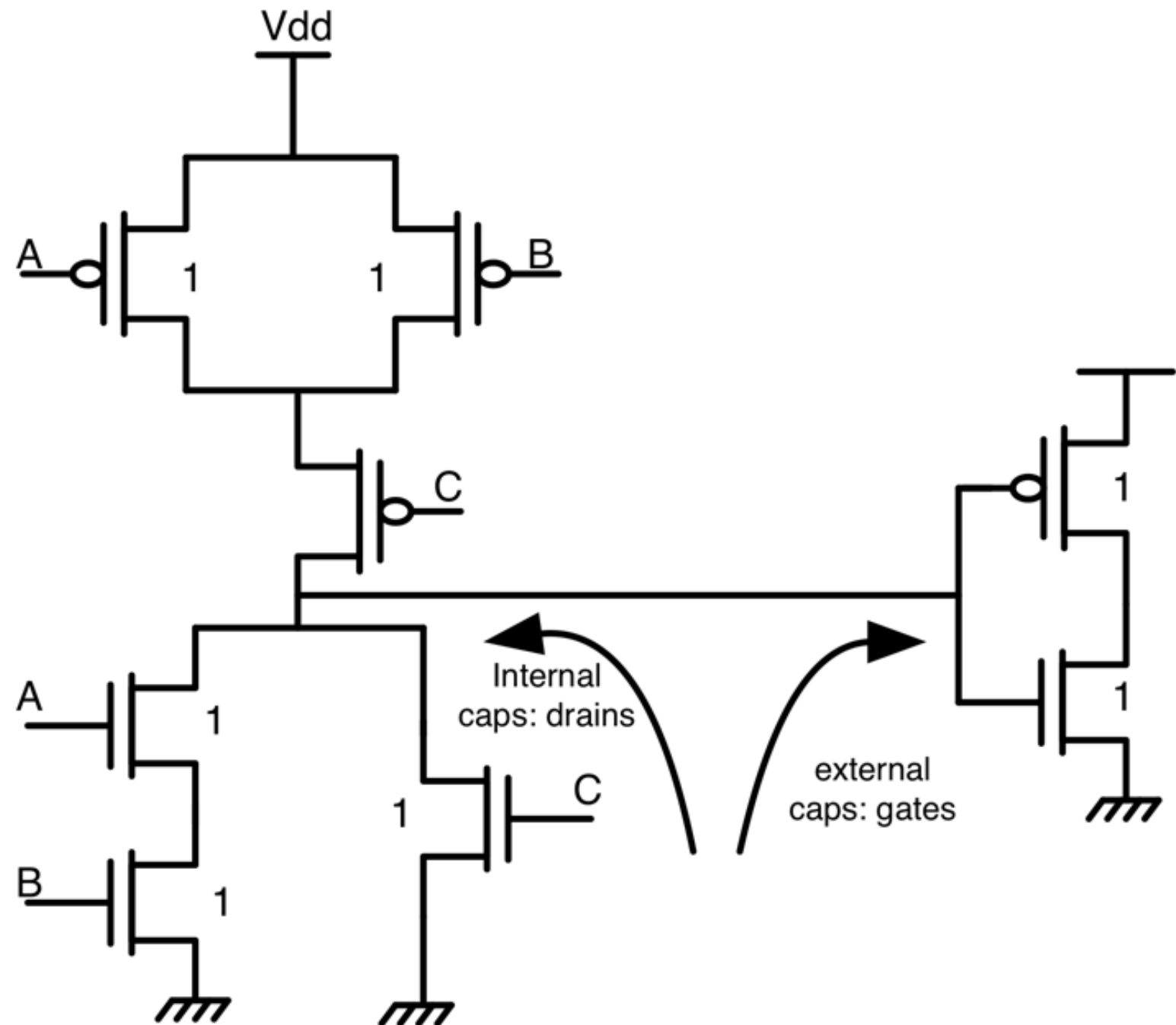
Logic function	Implementation through expression of F'				Implementation through expression of F			
	Main gate	Input Inverters	Output Inverters	Total	Main gate	Input Inverters	Output Inverters	Total
$F=AB$	4	4	0	8	4	0	2	6
$F=A'B'+C$	6	2	0	8	6	4	2	12
$F=(AB)'+CD$	8	4	0	12	8	4	2	14
$F=ABC+D'$	8	6	0	14	8	2	2	12
$F=A'+A'B$	2	0	0	2	2	2	0	2
$F=(ABC)'+(DE)'$	10	0	0	10	10	10	2	22

THE EFFECT OF TRANSISTOR SIZES

- Ratioless logic, transistor sizes have no impact on output values
- Transistor sizing up raise capacitance
 - Raises self loading through drains
 - Raises external loading on previous stage through gates
- Transistor sizing up reduces resistance
- Transistor size is the determining factor for delay

MINIMUM AREA GATE

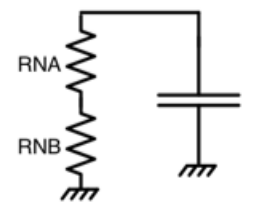
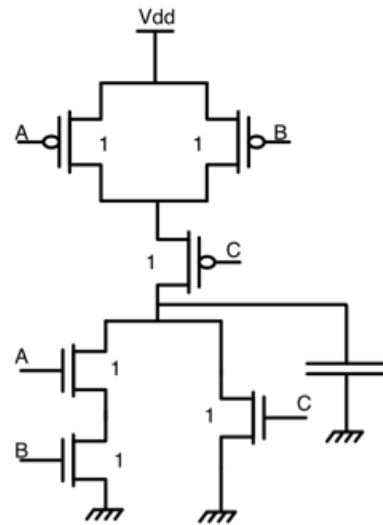
- Minimum area gate has minimum size transistors
- Multiple values for delay depending on activated branch



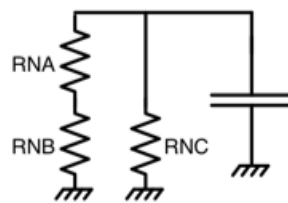
DELAY FOR DIFFERENT INPUT COMBINATIONS

.....

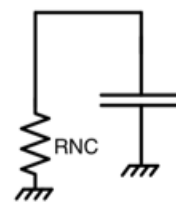
- Capacitance is always the same



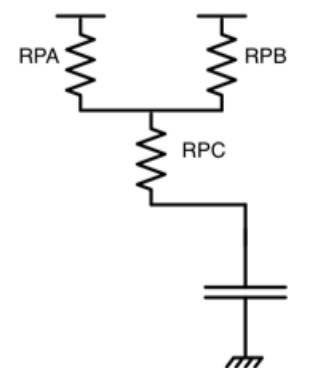
$$\begin{aligned} \text{ABC} &= '110' \\ R &= R_{NA} + R_{NB} \\ G &= G_{NA} * G_{NB} / (G_{NA} + G_{NB}) \end{aligned}$$



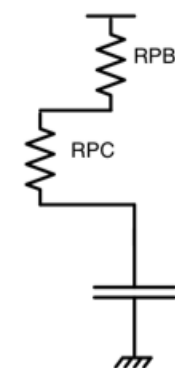
$$\begin{aligned} \text{ABC} &= '111' \\ R &= (R_{NA} + R_{NB}) \parallel R_{NC} \\ G &= (G_{NA} \parallel G_{NB}) + G_{NC} \end{aligned}$$



$$\begin{aligned} \text{ABC} &= '001' \\ R &= R_{NC} \\ G &= G_{NC} \end{aligned}$$



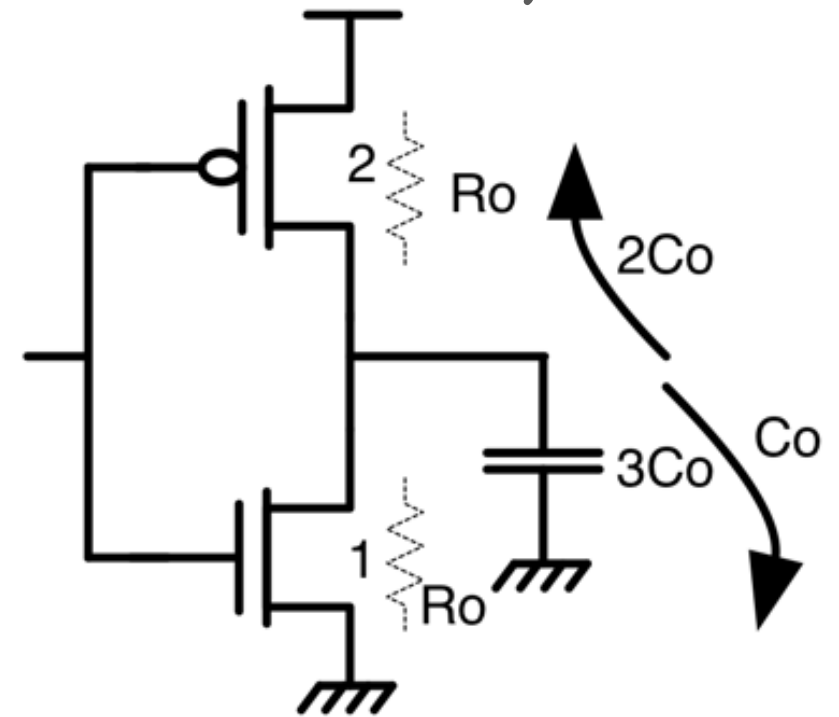
$$\begin{aligned} \text{ABC} &= '000' \\ R &= R_{PC} + (R_{PA} \parallel R_{PB}) \\ G &= R_{PC} \parallel (R_{PA} + R_{PB}) \end{aligned}$$



$$\text{ABC} = '100'$$

REFERENCE POINT

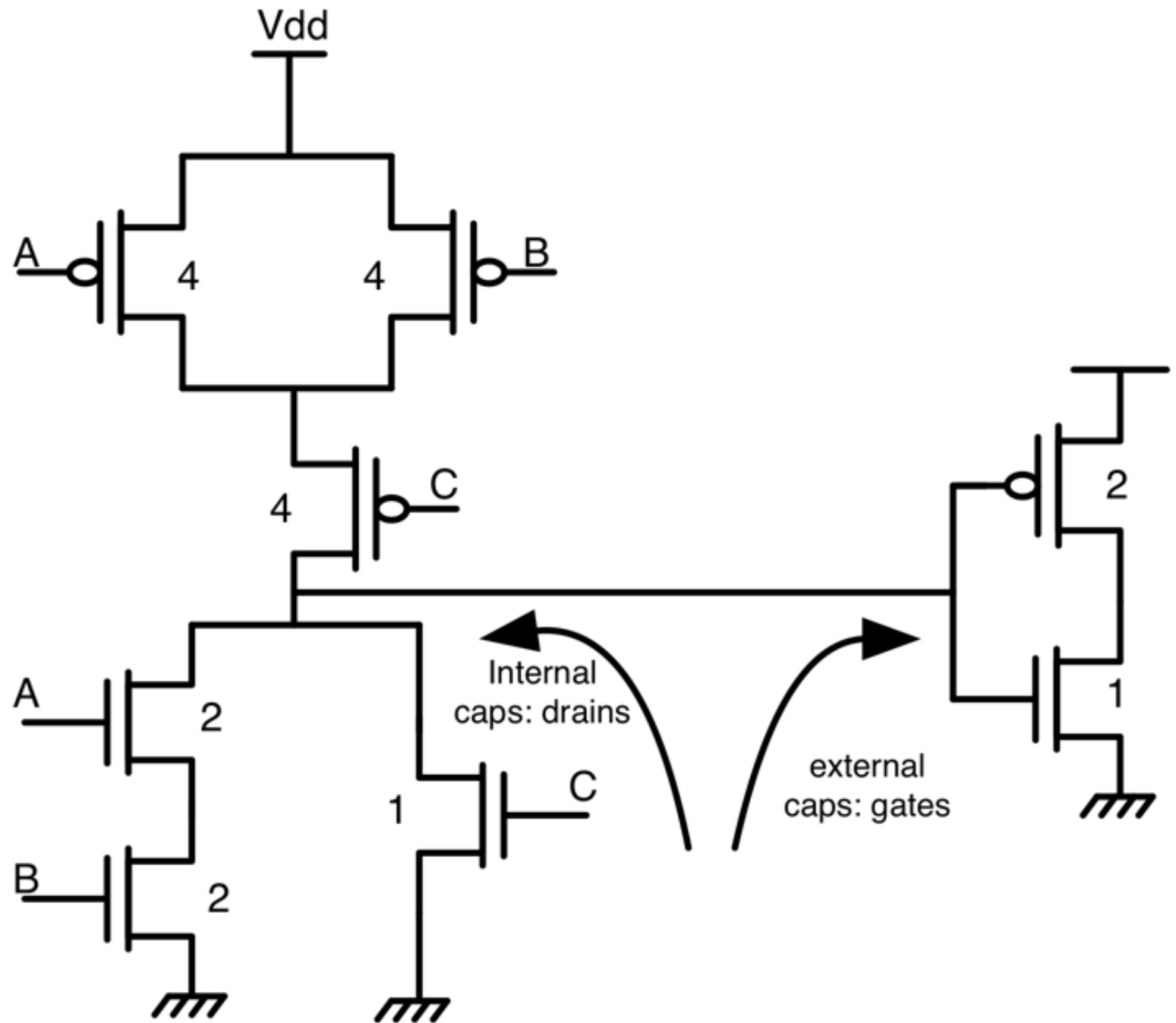
- Unit inverter
- Minimum size inverter for which HL and LH delays are equal



PROPER DESIGN APPROACH

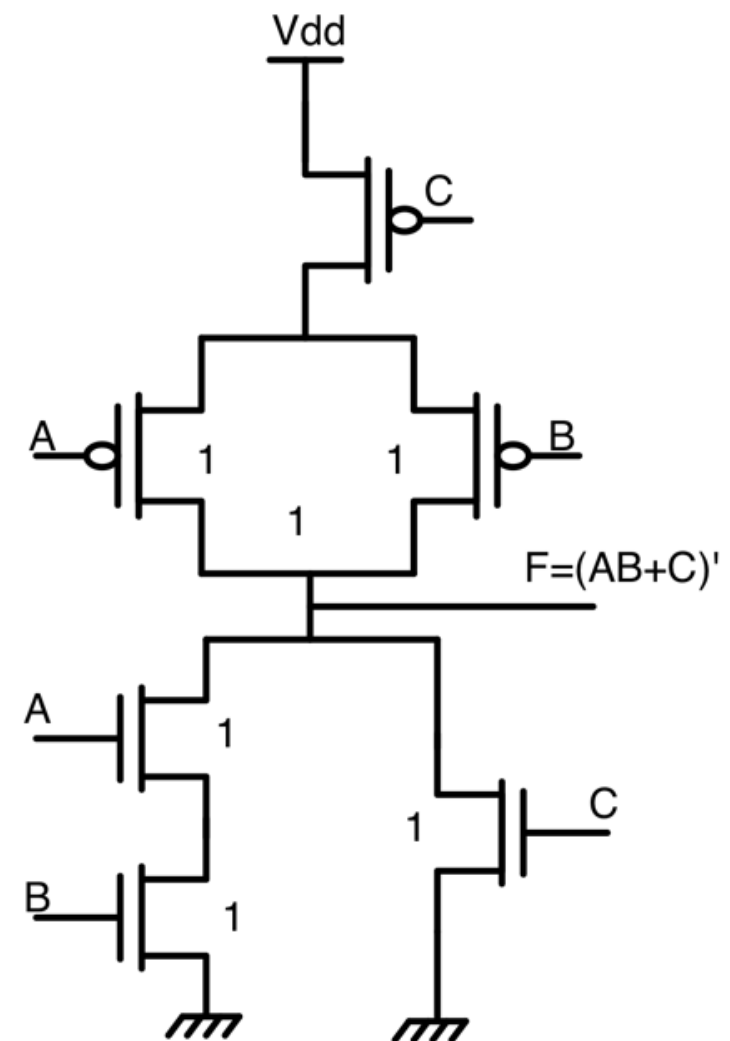
- Refer all resistances to the resistance of the unit inverter
- Size to achieve a certain worst case resistance
 - Why design for worst case?
 - What does worst case entail for CMOS circuits?

- Gate sized to guarantee worst case resistances equal to that of the unit inverter



FUNCTIONAL VS CORRECT

- This is a functional implementation of $F = (AB + C)'$
- It is not “correct”

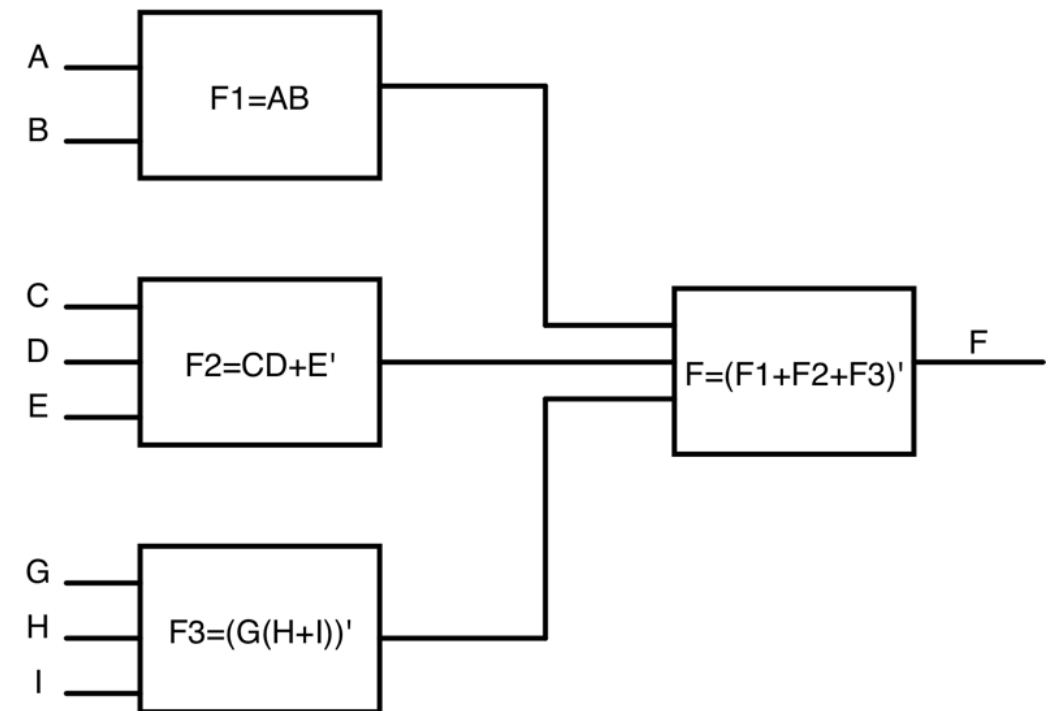


COMPOSITE IMPLEMENTATIONS

- So far logic function $\rightarrow$ logic gate
- A very large function may be implemented as multiple gates in parallel/cascade
- Proper level of granularity has to be determined on a case by case basis

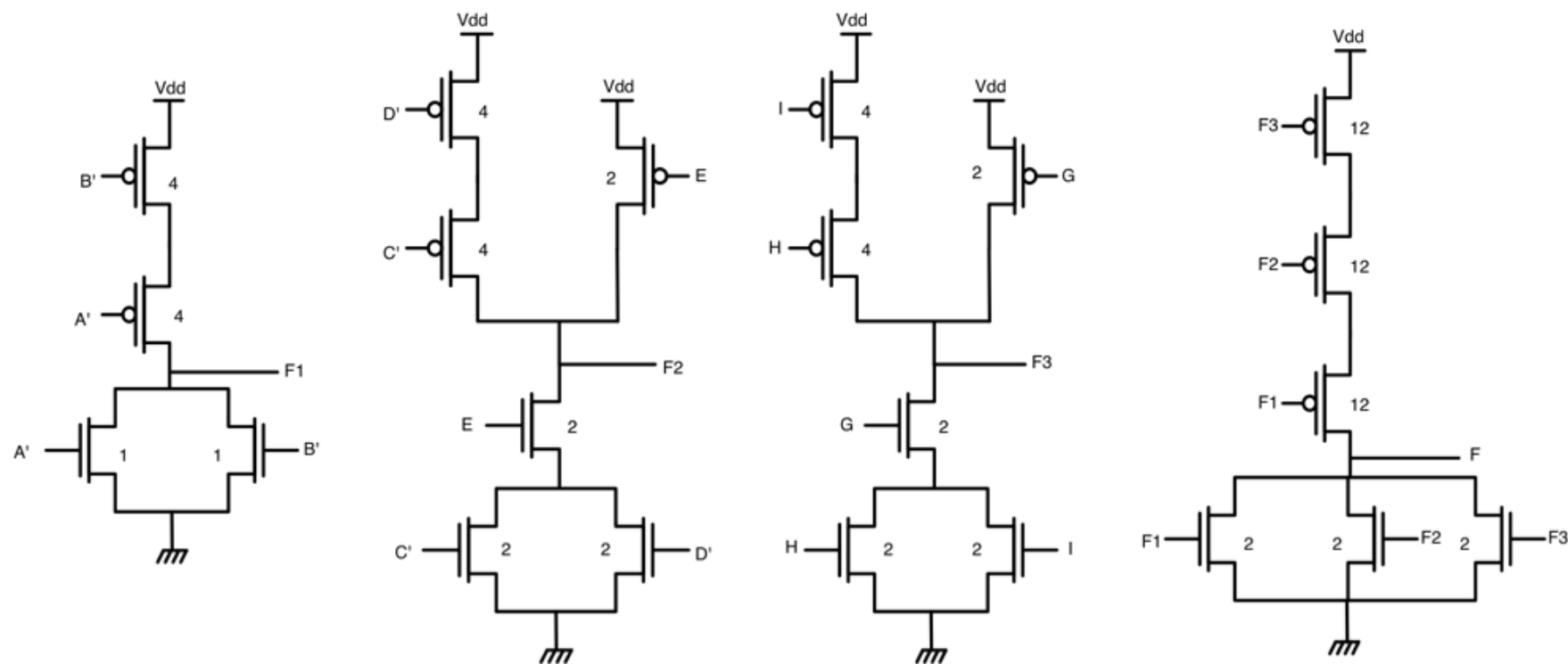
COMPOSITE IMPLEMENTATION

- Four gates
- One function

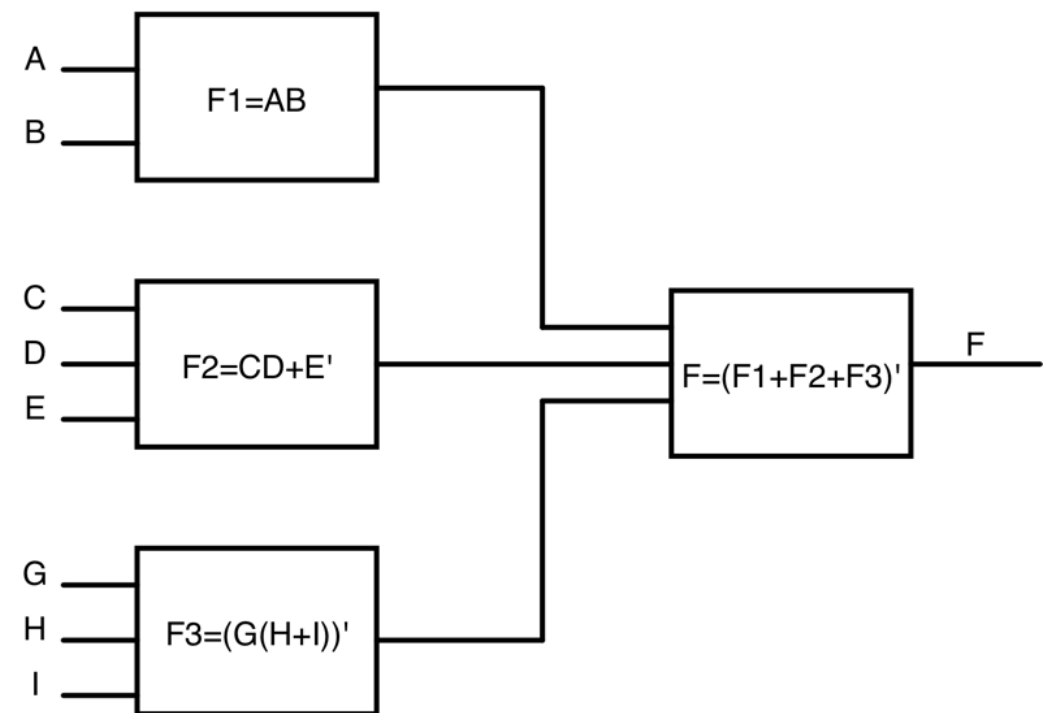


THE FOUR GATES

- All are sized for worst case resistance equal to unit inverter



-
- Delay expression is very topology dependent
 - But is immediately obvious

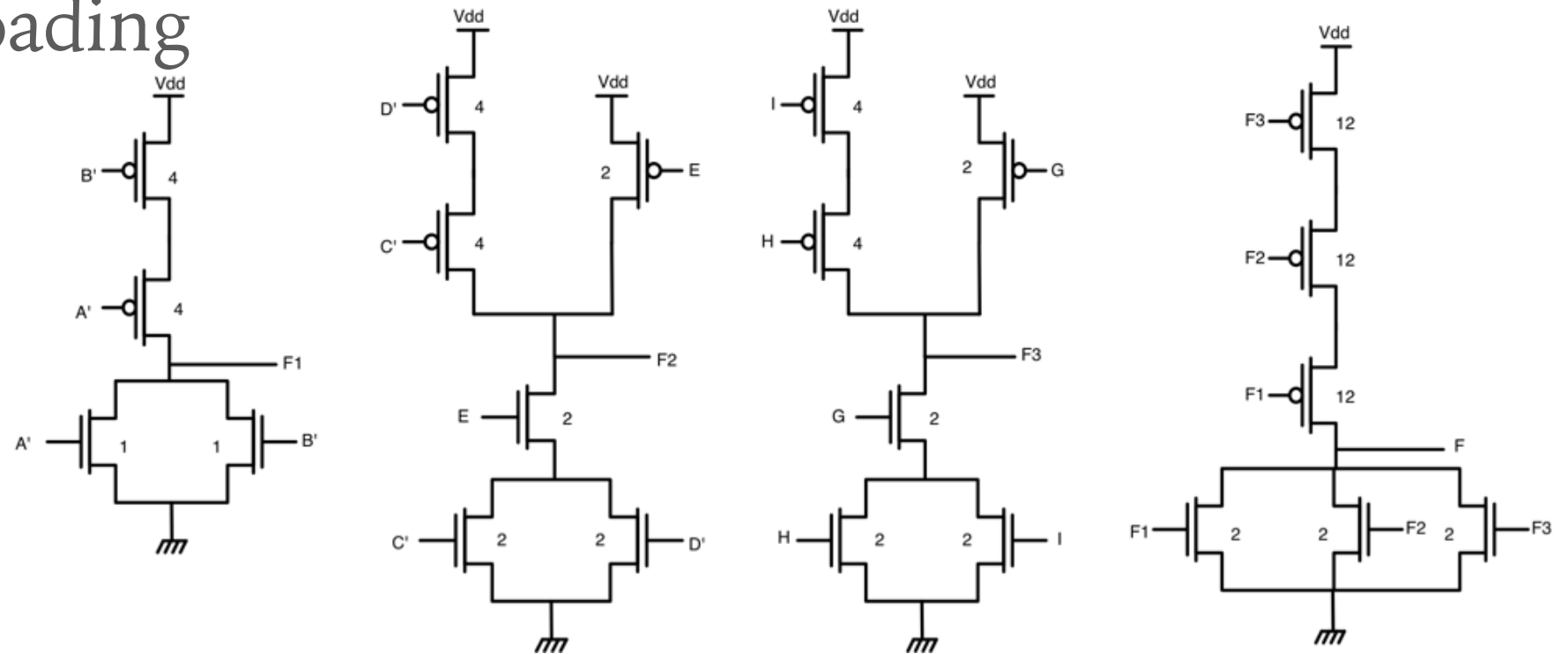


$$t_{phl} = \max\{t_{F1lh}, t_{F2lh}, t_{F3lh}\} + t_{Fhl}$$

$$t_{plh} = \max\{t_{F1hl}, t_{F2hl}, t_{F3hl}\} + t_{Flh}$$

ESTIMATING DELAYS

- First estimate loading
- Then delays



Gate	Total Output Capacitance	Of which self-	Worst case pull up W/L	Worst case pull down	High to low time	Low to high time
F1	20Co	6Co	2	1	20RoCo	20RoCo
F2	22Co	8Co	2	1	22RoCo	22RoCo
F3	22Co	8Co	2	1	22RoCo	22RoCo
F	18Co	18Co	4	2	9RoCo	9RoCo

A (VERY VERY) ROUGH ESTIMATE OF AREA

- This is only valid for preliminary comparisons
- Need only know aspect ratios

$$Area \approx \sum WL = \sum WL_{\min} = L_{\min} \sum W = L_{\min}^2 \sum \frac{W}{L}$$

EXPECTATIONS

- Raising channel width leads to an increase in area
 - We should ostensibly see an improvement in speed
- Raising supply increases power dissipation
 - We should again see an increase in speed
- Should this be seen through the resistance or capacitance part of the time constant?

CAPACITANCE OR RESISTANCE

- Capacitance is not a function of supply value
 - To a first order only, drain capacitance is a function of supply through C_j
- Resistance is the primary way in which delay is impacted by supply scaling

RESISTANCE AND SUPPLY

- Resistance is a diminishing function of supply
- This is especially clear for large values of supply

$$R_{eq} = \frac{3V_{DD}}{2\mu_n C_{ox}(W/L)_n (V_{DD} - V_{thn})^2}$$

$$R_{eq} = \frac{3}{2\mu_n C_{ox}(W/L)_n V_{DD}}$$

$$R_{eq} \propto \frac{1}{V_{DD}}$$

SHOULD WE SCALE SUPPLY UP?

- A higher supply will definitely lead to less delay
- However, power also rises
 - In fact, power rises disturbingly fast
 - It rises cubically
- The decision must be based on both metrics

RESTATING THE TIME CONSTANT

$$\tau = R_{gate}(C_{int} + C_{ext}) = R_{gate}C_{int} + R_{gate}C_{ext}$$

$$\tau = \tau_{int} + \tau_{ext}$$

- An internal (intrinsic, self) component
- And external component

SCALING WIDTH

- Raises self capacitance by S
- Reduces resistance by R

$$\tau = R_{gate}(C_{int} + C_{ext}) \xrightarrow{\text{scale by S}} \frac{R_{gate}}{S}(SC_{int} + C_{ext})$$

SHOULD WE SCALE UP TRANSISTORS?

.....

- If there is no external loading then definitely no

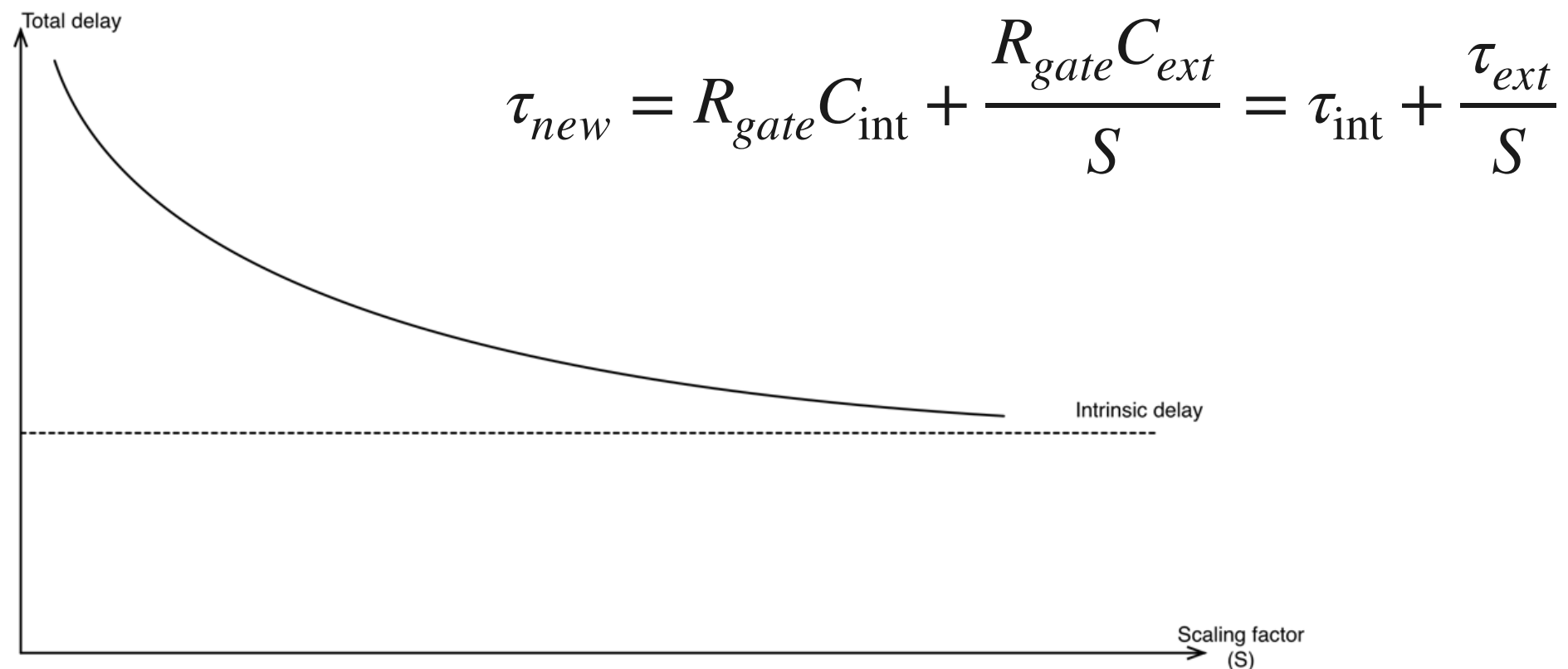


Figure. 3.10.1 graph of total delay versus gate scaling factor.

SHOULD WE SCALE UP TRANSISTORS

- If we have external loading
 - Scaling up transistors reduces external and thus total delay
 - The reduction is diminishing

$$\tau_{new} = R_{gate}C_{int} + \frac{R_{gate}C_{ext}}{S} = \tau_{int} + \frac{\tau_{ext}}{S}$$

- Delay can never drop below intrinsic delay
- But if we reduce our own delay, what happens to the stage before us?
- What is the correct answer?

RECAPPING THE ROAD TO CMOS

- Start by BJT logic
 - Dismal fanout and integration
- RTL logic
 - Large area to achieve low V_{ol}
- Enhancement load
 - Loss of headroom
- Depletion load and pseudo NMOS
 - Best possible ratioed performance, but fabrication difficulty
- CMOS
 - General goodness

ADVANTAGES OF CMOS

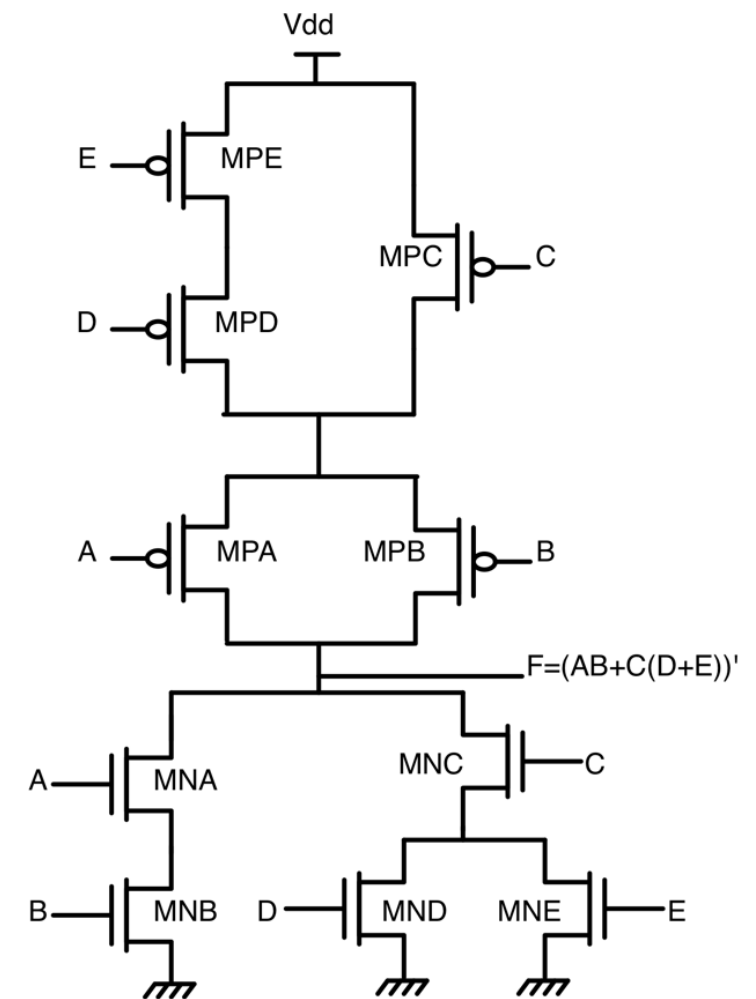
- Rail to rail logic
 - PMOS as a pull-up and NMOS as a pull-down switch
- Ratioless logic
 - Zero drop impedance cancels out all transistor parameters
- Zero static current
 - Thus zero static power and rail to rail logic
- Excellent noise margins
 - Marred only by channel length modulation

THE FIRST DISADVANTAGE

- The number of transistors is large compared to ratioed logic
- N in PDN, N in PUN
- CMOS $2N$ transistors
- Ratioed logic $N+1$ transistors
- Matters made worse in CMOS by need to size PMOS up to account for mobility

THE SECOND (AND REAL) DISADVANTAGE

- Each input feeds two transistor gates
- Loading on previous gate is “doubled”
- In fact it’s worse due to PMOS sizing
- Delay is almost doubled
- Sizing up or down does not help
 - Recall only chain delay matters



MOVING THE GOALPOSTS

- Our new “best” inverter should have the following:
 - Rail to rail logic
 - Ratioless
 - Zero static power
 - Large noise margins
 - Number of transistors $O(N)$
 - Loading by a single transistor gate
- If we manage to achieve this, we will certainly have to contend with a new set of problems